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Dear Editor, *PNAS*

We are pleased to submit our manuscript entitled "**Boundaries of Executive Functions in Large Language Models: GPT-4o Selectively Replicates Human Performance**" for consideration as a Research Article in the *Proceedings of the National Academy of Sciences*. This work has not been previously published nor is it under consideration elsewhere. All authors have reviewed the manuscript and agree to abide by the journal's ethical standards.

Large language models (LLMs) have demonstrated surprisingly human-like performance in complex reasoning and problem-solving tasks that traditionally depend on executive functions (EFs). However, the extent to which these models replicate fundamental patterns of human executive control remains poorly understood. Our study addresses this question through systematic evaluation of GPT-4o's performance across the established three-factor framework of EF—inhibitory control, working memory, and cognitive flexibility—using well-established behavioral paradigms from cognitive psychology. We also introduce log probability parameters as computational indicators to examine the internal processing mechanisms underlying LLM performance on these cognitive tasks (see figure below).

We tested GPT-4o using two independent datasets (N_1 = 1,970; N_2 = 39), simulating trial-by-trial responses while recording internal computational parameters. Our findings reveal a pattern of selective replication: the model successfully reproduces human-like performance in tasks involving symbolic representation and static information maintenance (such as interference inhibition and working memory capacity), while showing divergent patterns in tasks requiring dynamic control and temporal sensitivity (including response inhibition and cognitive flexibility). This selective correspondence indicates that different EF dimensions may involve distinct computational requirements—some may be accessible through language-based learning while others potentially depend on specialized mechanisms not present in current LLM architectures. Additionally, we examined log probability parameters as potential indicators of computational processing during cognitive task performance. These internal metrics show task-specific associations with behavioral performance and correspond with activation patterns in EF-related brain regions during working memory and task-switching. While exploratory, this methodological approach provides a quantitative framework for comparing artificial and biological cognitive processing.

Our work contributes to the field in several ways. Through systematic mapping of LLM performance across the complete EF framework, we identify specific boundaries in current LLMs' capacity to replicate human cognitive control. The observed selective replication patterns are consistent with computational heterogeneity within EF dimensions, raising the possibility that different aspects of executive control may have distinct computational bases. Furthermore, our multilevel analytical framework—combining behavioral, computational, and neural correspondence measures—offers new methodological tools for investigating cognitive capabilities in artificial systems.

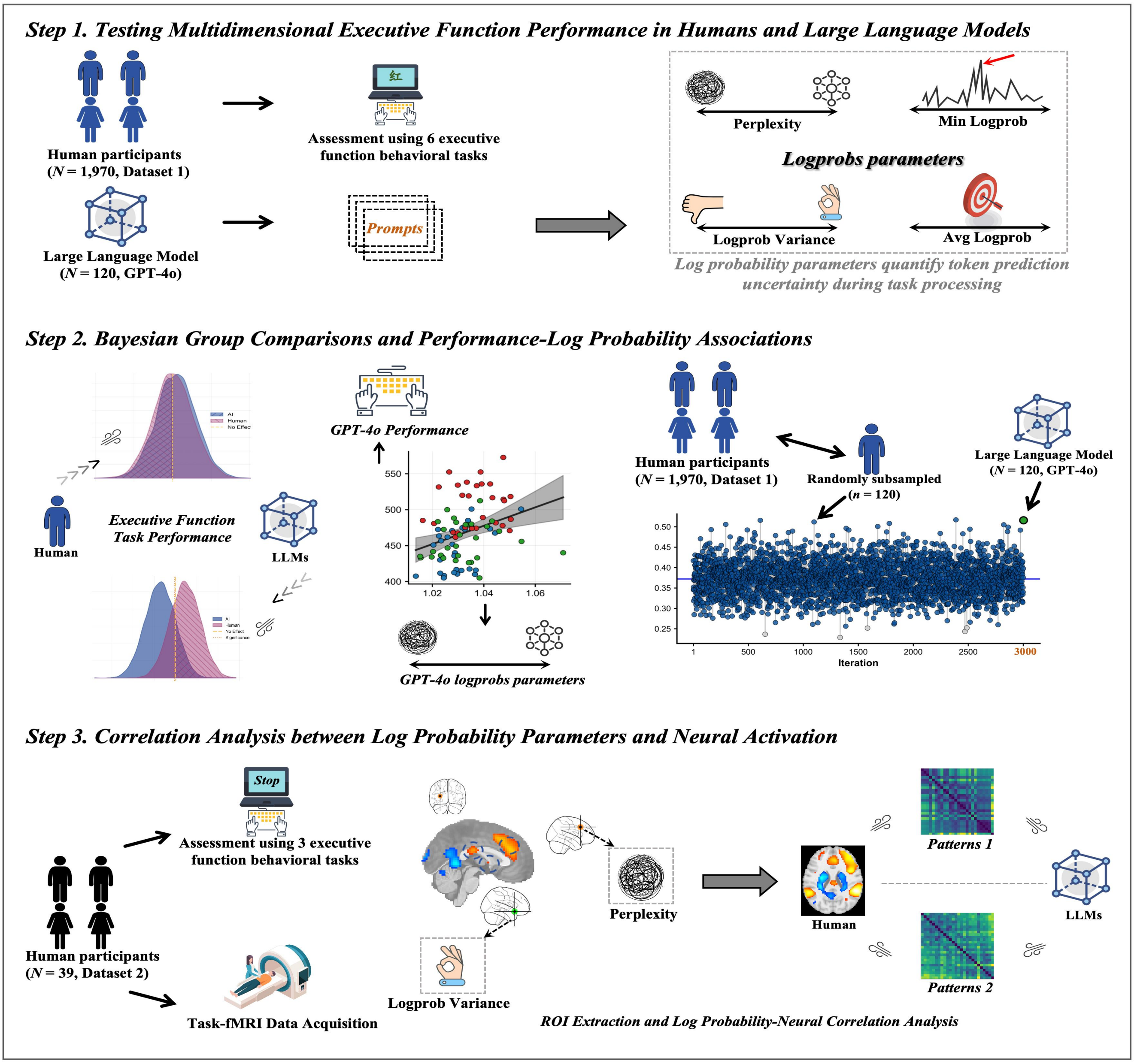
We believe this work will be of broad interest to *PNAS* readers across cognitive science, neuroscience, and artificial intelligence, as it provides empirical insights into the capabilities and limitations of LLMs in replicating functions associated with the prefrontal cortex, particularly executive functions.

Thank you for considering our manuscript.

Yours sincerely,

Xin Zhao and Jiang Qiu
On behalf of all authors

Study Overview



Suggested Reviewers

1. Antao Chen

- **Affiliation:** Brain Health Institute, National Center for Mental Disorders, Shanghai Mental Health Center, Shanghai Jiao Tong University School of Medicine and School of Psychology, Shanghai 200030, China
- **Email:** antao.chen@sjtu.edu.cn
- **Expertise:** Cognitive neuroscience, brain health, mental disorders

2. Xiaolin Zhou

- **Affiliation:** School of Psychology and Cognitive Science, East China Normal University, Shanghai, China
- **Email:** xz104@psy.ecnu.edu.cn
- **Expertise:** Cognitive psychology, attention and executive function

3. Joseph H.R. Maes

- **Affiliation:** Donders Institute for Brain, Cognition and Behaviour, Centre for Cognition, Radboud University, Nijmegen, The Netherlands
- **Email:** roald.maes@donders.ru.nl
- **Expertise:** Cognitive control, learning and memory

4. Shanka Subhra Mondal

- **Affiliation:** Princeton University, Princeton, NJ, USA
- **Email:** smondal@princeton.edu
- **Expertise:** Computational modeling, AI and cognition

5. Michal Kosinski

- **Affiliation:** Graduate School of Business, Stanford University, Stanford, CA, USA
- **Email:** michalk@stanford.edu
- **Expertise:** Computational social science, digital psychology, AI behavior

Boundaries of Executive Functions in Large Language Models: GPT-4o Selectively Replicates Human Performance

Tongyi Zhang [1], Jiang Qiu[2], and Xin Zhao[1]*
[1]School of Psychology, Northwest Normal University, Lanzhou, China
[2]Key Laboratory of Cognition and Personality of Ministry of Education, Faculty of Psychology, Southwest University

*Corresponding authors:
Xin Zhao ORCID: 0000-0001-9585-8306, Email: psyzhaoxin@nwnu.edu.cn

Author Notes

Classification: Biological Sciences / Psychological and Cognitive Sciences

Authors' contributions: T.Z. conceived and designed the experiments, conducted the experiments, analyzed the data, and drafted and revised the manuscript. J.Q. provided validated samples with fMRI data resources and contributed to manuscript revision. X.Z. conceived and designed the experiments, supervised the study, contributed to participant recruitment and data collection, and drafted and revised the manuscript. All authors reviewed and approved the final manuscript.

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Data, Materials, and Software Availability. Transcripts and numerical coding of transcripts data have been deposited in OSF (https://osf.io/hfnt6).

Abstract

Large language models (LLMs) have demonstrated considerable capabilities in complex reasoning and problem-solving tasks that, in humans, depend on executive functions (EFs). However, the extent to which these models replicate human EF patterns remains unclear. We systematically evaluated GPT-4o's performance across three core EF dimensions—inhibitory control, working memory, and cognitive flexibility—using established behavioral paradigms. Additionally, we examined whether model-internal log probability parameters could serve as quantitative indicators of cognitive processing analogous to human neural activity. Using two independent datasets (N_1 =1,970; N_2 =39), we simulated trial-by-trial responses through GPT-4o while recording log probability metrics. Bayesian analyses revealed selective replication of human EF patterns. GPT-4o successfully reproduced human-like performance in interference inhibition (*stroop*), working memory capacity (*digit span*), and working memory updating (*n*-back). In contrast, the model showed divergent patterns in response inhibition (*Go/No-Go*), time-sensitive working memory updating (*running memory* task with presentation times of 1750 ms and 750 ms), and cognitive flexibility (n*umber-switching* task). Log probability parameters demonstrated task-specific associations with behavioral measures and corresponded with activation patterns in EF-related brain regions during working memory and task-switching paradigms. These findings suggest that GPT-4o captures specific aspects of human EF, particularly those involving symbolic representation and static information maintenance, while showing limitations in dynamic control and temporal processing. This selective replication pattern provides insights into both the computational basis of EF and the cognitive boundaries of current LLM architectures. Our results indicate that log probability parameters may offer a window into LLM cognitive processing, providing a methodological framework for evaluating artificial cognitive mechanisms.

Keywords: Large language models; Executive functions; GPT; Log probability; Prefrontal cortex

Significance

Executive functions (EFs)—the cognitive control processes that enable goal-directed behavior—are fundamental to human intelligence and have traditionally been associated with specialized neural architecture in the prefrontal cortex. The emergence of large language models capable of complex reasoning tasks raises important questions about which cognitive capacities can arise from statistical learning of language patterns. By systematically evaluating GPT-4o across established EF paradigms, we demonstrate that current LLMs selectively replicate specific dimensions of human cognitive control. The model successfully captures interference resolution and working memory maintenance while showing divergent patterns in dynamic inhibitory control and cognitive flexibility. This selective replication suggests that certain EFs may emerge from domain-general computational principles accessible through language-based learning, whereas others potentially depend on specialized mechanisms not present in current architectures. Additionally, we show that model-internal log probability parameters correspond with both behavioral performance and human brain activation patterns, providing a quantitative framework for comparing artificial and biological cognitive processing. These findings contribute to our understanding of the computational basis of EFs and offer practical methods for evaluating cognitive capabilities in artificial systems, with implications for both cognitive science and AI development.

Introduction

Large language models (LLMs) have achieved remarkable performance on tasks traditionally used to assess human reasoning and problem-solving, including multi-step reasoning (1), strategic planning (2) and abstract concept manipulation (3). This convergence between machine and human performance raises fundamental questions about the nature of cognition: Can statistical learning systems develop processing mechanisms analogous to those underlying human cognitive functions? This question bears directly on our understanding of cognitive computation and suggests new methodological approaches for cognitive science (4–8). Through systematic behavioral comparisons between LLMs and humans, we can begin to identify which cognitive capacities arise from large-scale language learning and which may depend on biological or embodied constraints unique to human cognition (9, 10).

Executive functions (EFs) provide a particularly informative test case, as they constitute the cognitive control system that coordinates goal-directed behavior and enables complex reasoning and decision-making (11–13). Initial investigations of LLM performance on EF tasks have yielded mixed results (details in SI Appendix, Table S1). For working memory, LLMs display capacity constraints similar to those observed in humans (14, 15), though performance varies considerably across models and appears more dependent on training scale than architecture (16). The picture becomes more complex for cognitive flexibility and inhibitory control, where LLMs consistently struggle with dynamic rule adaptation, conflict resolution, and prospective planning—falling well below human benchmarks (2, 17, 18). Furthermore, while LLMs can reproduce certain psychological phenomena, they exhibit distinct patterns of cognitive biases and capabilities that do not straightforwardly improve with model scale (17, 19, 20). These findings suggest selective rather than general correspondence between LLM and human EF.

The multidimensional nature of EF presents both challenges and opportunities for this investigation. Following the influential framework of Miyake et al. (2000), EF comprises three core components—inhibitory control, working memory, and cognitive flexibility—that are separable yet interrelated (21). This tripartite structure is supported by converging neuroimaging evidence linking working memory with dorsolateral prefrontal cortex (22), conflict monitoring to anterior cingulate cortex (23), and response inhibition to inferior frontal gyrus (24) Despite this well-established framework, previous studies have typically examined LLM performance on individual executive tasks rather than systematically evaluating the full tripartite structure. This piecemeal approach leaves open whether LLMs exhibit selective strengths and weaknesses across executive domains or demonstrate more uniform capabilities. Mapping these patterns would clarify which EFs can emerge from statistical learning alone versus those requiring specialized cognitive architecture (25).

A key challenge for current research is its focus on behavioral outcomes without examining the computational processes that generate them. Establishing meaningful parallels between LLM and human cognition requires moving beyond surface-level behavioral similarities to understand underlying mechanisms. While cognitive neuroscience infers processing dynamics from neural activity patterns, comparable methods for examining LLM internal states remain underdeveloped. The log probability (logprobs) parameters available in OpenAI's GPT models offer one promising approach (26). These metrics—including perplexity, probability variance, average log probability, and minimum log probability—quantify the model's uncertainty and computational complexity during response generation (27–29). Systematic patterns in these parameters may reveal processing demands analogous to cognitive load in humans, potentially bridging behavioral and mechanistic levels of analysis. However, the relationship between these internal metrics and cognitive processes remains largely unexplored.

The present study addresses these gaps by conducting a comprehensive evaluation of GPT-4o's performance across the three-factor EF framework when presented with standard cognitive tasks as in human studies, testing whether current LLMs exhibit patterns consistent with established cognitive phenomena in inhibitory control, working memory, and cognitive flexibility (Fig. 1). Beyond behavioral assessment, we analyze log probability parameters to characterize the model's internal processing dynamics and examine their correspondence with human neural activity patterns from task-functional magnetic resonance imaging (fMRI) studies.

Methods

Participants

We analyzed two independent datasets to investigate EF performance (SI Appendix, Table S2). Dataset 1 consisted of behavioral data collected for this study from 1,970 native Mandarin-speaking adults (915 males, 46.45%; 1,055 females, 53.55%; mean age = 19.78 years, SD = 1.72) who completed five standardized EF tasks. Participants were recruited from Northwest Normal University and surrounding communities. Inclusion criteria required native Mandarin proficiency, normal or corrected-to-normal vision, and normal color vision. Individuals with neurological or psychiatric disorders or those taking medications affecting cognitive function were excluded. Dataset 2 consisted of fMRI data from He et al. (2021), including 39 adults (13 males, 26 females; mean age = 19.21 ± 0.52 years) who performed three EF tasks during scanning (30). All human participants provided written informed consent, and the study was approved by the Ethics Committee of Northwest Normal University (Protocol #NUNU-20250105).

Testing human EF performance

Participants in Dataset 1 completed five EF tasks using E-Prime software (SI Appendix, Fig. S1, for details, see SI Appendix, Methods). The battery evaluated core EF components: (1) The *Stroop color-word* task assessed interference control by requiring participants to identify ink colors while ignoring word meanings across congruent, incongruent, and neutral conditions (18 practice trials, 108 test trials). (2) The *Go/No-Go* task measured response inhibition, with participants responding to target letters while withholding responses to non-targets in a counterbalanced design (4 blocks × 100 trials). (3) The *number switching* task evaluated cognitive flexibility through alternating magnitude judgments for red digits and parity judgments for blue digits across single-task blocks (10 blocks × 8 trials) and mixed-task blocks (10 blocks × 17 trials). (4) The *running memory* task assessed working memory updating by requiring continuous updating of the last three digits under simple (1750 ms presentation) and difficult (750 ms) conditions across sequences of varying lengths (5-11 items). (5) The *digit span* tasks measured working memory maintenance through forward and backward recall using adaptive procedures beginning with three-digit sequences.

Dataset 2 participants performed three tasks during fMRI scanning: (1) The *n*-back task assessed working memory across 0-back, 1-back, and 2-back conditions using blocked designs (4 blocks per condition, 29s each with 15 stimuli). (2) The *stop-signal* task evaluated response inhibition with 96 go trials and 32 stop trials, employing dynamically adjusted stop-signal delays. (3) The *number-letter switching* task measured cognitive flexibility with 48 repeat and 48 switch trials, requiring alternation between parity and consonant/vowel judgments based on color cues.

Testing LLM EF Performance and Log Parameters

We evaluated GPT-4o (version 2025-03-26) through OpenAI's official Application Programming Interface (API) with temperature set to 0 to ensure consistency with prior LLM cognitive research (19, 20, 31–33), facilitating direct comparison of results across studies. To verify the robustness of our findings, we repeated all analyses at temperature = 1 as a sensitivity check (20, 31). All other parameters used default settings. The model completed EF tasks through prompt-driven multi-turn conversations, with each simulated participant maintaining consistent context within a single dialogue session to ensure ecological validity and independence between participants. Task prompts and procedures closely matched those used with human participants (34), delivered to GPT-4o through natural language inputs (details in SI Appendix, Table S3-4). To quantify internal computational processes (SI Appendix, Table S5), we extracted four log probability parameters during response generation (see https://cookbook.openai.com/examples/using_logprobs):

(1) Average log probability quantified overall model confidence:

$$\text{avg\_logprob} = \frac{1}{n}\sum_{i=1}^{n} \log p\left(t_i\right)$$

(2) Perplexity measured model uncertainty:

$$\text{PPL} = \exp\left(-\text{avg\_logprob}\right)$$

(3) Log probability variance indicated confidence variability:

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n} \left(\log p\left(t_i\right) - \text{avg\_logprob}\right)^2$$

(4) Minimum log probability identified weakest predictions:

$$\text{min\_logprob} = \min_{i=1,\dots,n} \log p\left(t_i\right)$$

All computations were implemented using *NumPy* (35).

MRI data acquisition and analysis

Functional MRI data were acquired using a 3T Siemens scanner with echo-planar imaging sequences (TR = 2000ms, TE = 30ms, voxel size = $3.4\times3.4\times3\text{mm}^3$). High-resolution T1-weighted structural images were obtained using Magnetization Prepared Rapid Gradient Echo (MPRAGE) sequences. Preprocessing followed standard procedures using Statistical Parametric Mapping 12 (SPM12) and Data Processing & Analysis for Brain Imaging (DPABI) toolboxes (36), including slice-timing correction, realignment, coregistration, Diffeomorphic Anatomical Registration Through Exponentiated Lie Algebra (DARTEL) segmentation, normalization to Montreal Neurological Institute (MNI) space, and spatial smoothing with an 8mm Full Width at Half Maximum (FWHM) Gaussian kernel. Task-based activation was modeled using general linear models with the following contrasts: "2-back > 0-back" for working memory load, "successful stop > successful go" for inhibitory control, and "switch > repeat" for cognitive flexibility costs. Mean activation values were extracted from 116 anatomically-defined regions of interest using the Automated Anatomical Labeling (AAL) atlas (37) (see SI Appendix, Methods).

Sample size justification and considerations

Sample size determination accounted for fundamental differences between human and LLM response distributions. While human samples reflect genuine individual differences in EF, LLMs with fixed temperature parameters generate statistical variation around model-determined central tendencies (20). Given the more homogeneous response patterns of LLMs, excessive sampling risks pseudo-precision and violates the value of information principle (38, 39). We generated 120 simulated participants for Dataset 1, providing > 90% power to detect medium effects ($d = 0.5$) and >99% power for large effects ($d = 0.8$) at $\alpha = 0.05$, consistent with typical EF effect sizes ($d = 0.4$-0.8)(40). To ensure fair comparisons and avoid statistical biases from unequal sample sizes, we randomly subsampled 120 participants from the original 1,970 human participants for matched analyses. For Dataset 2, we generated 39 simulated participants to match the fMRI sample size, enabling direct comparison of internal computational parameters with neural activation patterns.

Data Analysis

Bayesian Analysis of Variance (ANOVA) was used to compare EF performance (SI Appendix, Table S6) between AI and humans (41), quantifying evidence strength through Bayes Factors (BF) (42). Analyses were implemented in Python using Python Monte Carlo (PyMC) with weakly informative priors (43). We computed posterior distributions for effect size differences, reporting posterior means and medians as point estimates along with 95% Highest Density Intervals (HDIs) to characterize uncertainty. BF quantified support for the hypothesis that AI effects exceed human effects:

$$BF_{10} = \frac{P(\Delta_{\text{effect}} > 0|\text{data})}{P(\Delta_{\text{effect}} \leq 0|\text{data})}$$

Following established conventions (44), $BF_{10} > 10$ indicated strong evidence, $3 < BF_{10} < 10$ indicated moderate evidence, $1 < BF_{10} < 3$ indicated weak evidence, and $BF_{10} < 0.33$ supported the null hypothesis.

To investigate relationships between internal computational mechanisms and behavioral performance, we performed Pearson correlation analyses between task performance metrics (reaction times, accuracy) and log probability parameters. All correlation coefficients were corrected for multiple comparisons using the Benjamini-Hochberg False Discovery Rate (FDR) procedure at $\alpha = 0.05$ (45).

Data, Materials, and Software Availability

Transcripts and numerical coding of transcripts data have been deposited in OSF (https://osf.io/hfnt6).

Results

Selective Replication of Human EF Effects in GPT-4o

EF Dimension I: Inhibitory Control

In the *Stroop* task, human participants demonstrated the typical interference effect. Specifically, one-way ANOVA revealed significant differences in reaction times across the three conditions ($p = 0.001$) (Fig. 2*A*). Post-hoc comparisons showed no significant difference between congruent and neutral conditions (p = 0.363), while reaction times in the incongruent condition were significantly longer than both congruent (p < 0.001) and neutral conditions (p = 0.009), demonstrating the classic Stroop interference effect. Similarly, GPT-4o exhibited a comparable interference pattern on the same task. One-way ANOVA showed significant differences in reaction times across conditions (p < 0.001) (Fig. 2*A*). Specifically, reaction times in the incongruent condition were significantly longer than both congruent (p < 0.001) and neutral conditions (p = 0.002), with no significant difference between congruent and neutral conditions (p = 0.384). Notably, Bayesian analysis revealed that GPT-4o's mean interference effect (89.210 ms) exceeded that of humans (51.570 ms), with a 95% HDI of [26.160, 52.620] ms and BF_{10} > 100. Moreover, analysis of GPT-4o's output probability distributions revealed significant differences in perplexity across conditions (p < 0.001), with incongruent trials showing higher perplexity than both congruent (p < 0.001) and neutral conditions (p < 0.001) (Fig. 2*A*). Similar patterns emerged for log probability variance (p < 0.001), average log probability (p < 0.001), and minimum log probability (p < 0.001). Furthermore, correlation analyses showed that GPT-4o's reaction times positively correlated with perplexity (r = 0.306, p_{fdr} < 0.001) and log probability variance (r = 0.280, p_{fdr} < 0.001), while negatively correlating with average log probability (r = −0.314, pfdr < 0.001) and minimum log probability (r = -0.272, p_{fdr} < 0.001) (Fig. 2*C*). Condition-specific analyses revealed that in congruent trials, reaction times correlated with perplexity (r = 0.339, p < 0.001) and log probability variance (r = 0.404, p_{fdr} < 0.001); in incongruent trials, reaction times correlated with perplexity (r = 0.358, p_{fdr} < 0.001) and average log probability (r = −0.383, p_{fdr} < 0.001); while neutral conditions showed no significant correlations (all p_{fdr} > 0.050).

Similarly, in the *Go/No-Go* task, human participants exhibited typical response inhibition difficulties, with significantly lower accuracy on No-Go trials compared to Go trials (p < 0.001) (Fig. 2*B*). In contrast, GPT-4o achieved 100% accuracy on both trial types. Consequently, Bayesian analysis showed a posterior mean difference in inhibitory control effects of 0.110, with a 95% HDI of [0.086, 0.133] and BF_{10} > 100. Despite perfect behavioral accuracy, GPT-4o's internal processing states revealed different patterns. Specifically, No-Go trials showed significantly higher perplexity than Go trials (p < 0.001). This difference was also evident in log probability variance (p < 0.001), average log probability (p < 0.001), and minimum log probability (p < 0.001) (Fig. 2*B*).

EF Dimension II: Working Memory

In the *digit span* task, human participants demonstrated classic working memory capacity characteristics, with forward digit span significantly exceeding backward digit span (p < 0.001) (Fig. 3*A*). Similarly, GPT-4o showed the same effect pattern (p < 0.001), though its digit span capacity exceeded that of human participants (−2.830, 95% HDI [-3.660, -1.960], BF_{10} approaching 0). Furthermore, analysis of GPT-4o's internal processing states revealed significant differences in perplexity between conditions ($p < 0.001$), with backward tasks showing higher perplexity than forward tasks. Additionally, log probability variance also differed significantly ($p < 0.001$), with backward tasks showing greater variance. Both average log probability ($p < 0.001$) and minimum log probability ($p < 0.001$) indicated lower predictive values in backward tasks (Fig. 3*A*). Moreover, correlation analyses showed that in forward tasks, sequence length correlated with perplexity ($r = 0.306$, $p_{fdr} < 0.001$), negatively with average log probability ($r = -0.307$, $p_{fdr} < 0.001$), and negatively with minimum log probability ($r = -0.313$, $p_{fdr} < 0.001$). In backward tasks, correlations were stronger: sequence length with perplexity ($r = 0.601$, $p_{fdr} < 0.001$), with average log probability ($r = -0.594$, $p_{fdr} < 0.001$), and with minimum log probability ($r = -0.597$, $p_{fdr} < 0.001$) (Fig. 3*C*).

In contrast, in the *running memory* task, human participants showed a refresh time effect, with higher average accuracy under 1750 ms stimulus presentation compared to 750 ms ($p = 0.001$) (Fig. 3*B*). However, GPT-4o showed no significant difference between conditions ($p = 0.833$). Accordingly, Bayesian analysis revealed an effect difference of -0.012, 95% HDI [-0.049, 0.020], $BF_{10} = 3.090$. Nevertheless, analysis of GPT-4o's internal processing states showed significant differences in perplexity between conditions ($p = 0.002$), with lower perplexity in the 750 ms condition compared to 1750 ms. Similarly, average log probability ($p = 0.002$) and minimum log probability ($p = 0.002$) also showed higher values in the 750 ms condition (Fig. 3*B*).

EF Dimension III: Cognitive Flexibility

In the *number switching* task, human participants demonstrated classic task-switching costs, with significantly longer reaction times in mixed-task switch trials compared to repeat trials (switch cost, $p < 0.001$), and longer reaction times in mixed-task repeat trials compared to single-task trials (mixing cost, $p < 0.001$). While GPT-4o showed significant switch costs ($p = 0.008$), these were smaller than humans (Fig. 4*A*). Indeed, Bayesian analysis revealed GPT-4o's switch costs were lower than humans' (−145.090 ms, 95% HDI [−202.210, -90.730] ms, BF_{10} approaching infinity). Additionally, GPT-4o's mixing cost was 58.190 ms. Between-condition comparisons showed no significant difference between mixed-task switch and repeat trials ($p = 0.958$), but significant differences between mixed-task conditions and single-task conditions (switch vs. single: $p = 0.008$; repeat vs. single: $p = 0.009$). However, analysis of GPT-4o's internal processing states revealed no significant between-condition differences in perplexity (all $p > 0.050$), with log probability variance and average log probability similarly showing no significant differences (Fig. 4*A*).

Replication in Independent Sample Across Three EF Dimensions

To validate our findings, we analyzed three EF tasks from Dataset 2. In the *n*-back task (Fig. 5*A*), LLMs demonstrated working memory load effects. Specifically, accuracy decreased with increasing *n*-back levels (0-back: 100%; 1-back: 93.2%; 2-back: 82.1%), with significant main effects ($p < 0.001$). Post-hoc comparisons revealed significant differences between all conditions: 0-back vs. 1-back ($d = 2.850$), 0-back vs. 2-back ($d = 2.340$), and 1-back vs. 2-back ($d = 1.380$). While reaction times increased with working memory load, this increase was not significant ($p = 0.228$). Overall, *n*-back effects showed an 17.910% accuracy decrease ($p < 0.001$). Additionally, internal state parameters varied with working memory load. Perplexity and log probability variance decreased with increasing load, with *n*-back effects of -0.011 and -0.005 respectively (both $p < 0.001$). Conversely, average and minimum log probabilities became more positive with increasing load (both $p < 0.001$). Furthermore, correlation analyses (Fig. 5*D*) showed model accuracy positively correlated with perplexity ($r = 0.529$, $p_{fdr} < 0.001$) and log probability variance ($r = 0.534$, $p_{fdr} < 0.001$), while negatively correlating with average log probability ($r = -0.533$, $p_{fdr} < 0.001$) and minimum log probability ($r = -0.587$, $p_{fdr} < 0.001$). In contrast, reaction times showed weak correlations with log probability parameters (all $p_{fdr} > 0.050$).

Similarly, in the *stop-signal* task (Fig. 5*B*), LLMs demonstrated response inhibition capabilities. Successful stop trials showed increased perplexity compared to successful go trials ($p < 0.001$, $d = 0.880$) and higher log probability variance ($p = 0.002$, $d = 0.690$). Correspondingly, average log probability decreased ($p < 0.001$, $d = -0.870$), and minimum log probability showed more negative values ($p < 0.001$, $d = -1.030$). Finally, in the *letter-number switching* task (Fig. 5*C*), LLMs did not show typical human switch cost patterns. The reaction time switch cost was −0.630 ms ($p = 0.228$, d = 0.022). However, internal state parameters revealed switch cost effects. Specifically, perplexity was higher in switch trials than repeat trials, producing a switch cost of 0.006 ($p < 0.001$, $d = 1.070$). Similarly, log probability variance, average log probability, and minimum log probability all showed significant switch costs (all $p < 0.001$, $d = 1.030$-1.110).

Potential Correspondence Between LLM Log Probability Parameters and Human Brain Activation

After examining LLM behavioral performance, we explored whether internal computational parameters might correspond to human neural activity patterns. In the *stop-signal* task "successful stop > successful go" contrast (Fig. 6*A*), we found no significant correlations between brain region activation and LLM internal computational parameters. However, in the *n*-back working memory task "2-back > 0-back" contrast (Fig. 6*B*), we discovered brain-model correspondences. Specifically, left gyrus rectus activation positively correlated with model perplexity ($r = 0.340$, $p_{fdr} = 0.037$) and negatively with average log probability ($r = -0.338$, $p_{fdr} = 0.038$). Similarly, the right medial superior frontal gyrus showed comparable patterns, with activation positively correlating with perplexity ($r = 0.328$, $p_{fdr} = 0.044$) and negatively with average log probability ($r = -0.325$, $p_{fdr} = 0.046$). Furthermore, in the *task-switching* paradigm's switch cost analysis (Fig. 6*C*), left thalamus activation positively correlated with model minimum log probability ($r = 0.323$, $p_{fdr} = 0.048$) and negatively with perplexity ($r = -0.322$, $p_{fdr} = 0.048$).

Sensitivity Analysis in Subsampling and Different Temperature Settings

Subsampling Analysis: To assess the robustness of our findings and ensure valid comparisons between the large human sample ($N_1 = 1{,}970$) and the GPT-4o responses ($N = 120$), we conducted bootstrap resampling analyses (Fig. 4*B*). We randomly sampled 120 human participants with replacement across 3,000 iterations and compared each subsample with the GPT-4o data. For the *Stroop* Task, 99.8% of human subsamples showed significant interference effects (mean $d = 0.372$, 95% CI [0.371, 0.374]). GPT-4o demonstrated a larger effect ($d = 0.516$), with all iterations confirming this difference ($BF_{10} \geq 10$). For the *Go/No-Go* Task, human participants consistently showed inhibitory control difficulties (mean $d = -0.866$, 95% CI [-0.870, -0.861]), whereas GPT-4o achieved near-perfect accuracy ($d < 0.001$; $BF_{10} > 100$ across all iterations). In the *Digit Span* Task, both groups showed forward-backward span differences (human: $d = -0.701$, 95% CI [-0.705, -0.697]; GPT-4o: $d = -2.238$), with 97% of iterations revealing qualitatively similar patterns despite magnitude differences ($BF_{10} < 0.1$). For the *Working Memory Updating* Task, only 37.8% of human subsamples showed time-dependent effects (mean $d = -0.226$, 95% CI [-0.230, -0.223]), whereas GPT-4o showed no such effect ($d = -0.027$, $p = 0.833$), with 98.5% of iterations supporting the null hypothesis (median $BF_{10} = 3.07$). In the *Task Switching* paradigm, all human subsamples showed robust switch costs (mean $d = 0.662$, 95% CI [0.659, 0.664]). GPT-4o exhibited smaller but significant costs ($d = 0.347$; $BF_{10} > 100$ across all iterations). These analyses confirm the stability of our findings and demonstrate the selectivity in GPT-4o's replication of human EF patterns.

Temperature parameter analysis: The temperature parameter in LLMs controls response variability, with lower values producing more deterministic outputs and higher values increasing stochasticity—a feature that may parallel individual differences in human cognitive processing (46). To examine whether our findings reflected stable patterns rather than artifacts of specific parameter settings, we repeated all analyses at temperature = 1. The temperature manipulation did not alter the main effect patterns across EF tasks, consistent with recent evidence showing minimal impact of temperature parameters on LLM performance in standardized psychological assessments (20, 31, 47). Specifically, in the *Stroop*, *Go/No-Go*, running memory, and *task-switching* paradigms, the direction of effects, statistical significance, and effect sizes remained consistent between temperature = 0 and temperature = 1 conditions (see SI Appendix, Fig. S2-4).

Discussion

This study systematically evaluated GPT-4o's performance when instructed to simulate human-like responses across behavioral tasks measuring three core dimensions of EF, while exploring the potential of model log probability parameters as indicators of cognitive processing. Our results revealed that GPT-4o successfully replicated typical human cognitive patterns in interference inhibition, working memory maintenance, and updating tasks. However, the model exhibited divergent patterns from humans in response inhibition, time-constrained working memory updating, and cognitive flexibility tasks. This selective replication pattern delineates specific boundaries in current LLM architectures when simulating human executive control mechanisms (SI Appendix, Fig. S5). Furthermore, log probability parameters demonstrated promise as cognitive processing assessment tools in certain task contexts, offering new empirical approaches for understanding the computational foundations of EFs.

Our findings demonstrate that GPT-4o, when prompted to simulate human-like responses, exhibited patterns consistent with human EF in tasks involving information representation conflicts and static information maintenance. In the *Stroop* task, the model displayed significantly prolonged reaction times in incongruent conditions, with effect sizes marginally exceeding typical human performance. This difference may reflect LLMs' nature as computational systems unaffected by fatigue or attentional fluctuations. These results align with Cui et al.'s (2025) observation that LLMs reliably replicate main effects while struggling with interaction effects (20). In working memory tasks, GPT-4o demonstrated structural constraints paralleling those of humans. The model replicated the backward difficulty effect in *digit span* tasks and exhibited performance degradation with increasing load in *n*-back tasks, suggesting that even when task comprehension is adequate, models exhibit structural working memory limitations (48). In contrast, GPT-4o failed to demonstrate typical human patterns in tasks requiring dynamic control and temporal sensitivity. The model achieved perfect accuracy in response inhibition tasks (*Go/No-Go* and *stop-signal*), showed no effects of time pressure in working memory updating tasks, and exhibited minimal switch costs in cognitive flexibility tasks. These findings converge with recent *Wisconsin Card Sorting Test* (WCST) studies, where Langis et al. (2025) and Goto et al. (2025) documented similar flexibility failures—the latter reporting that most LLMs achieved less than 30% accuracy with systematic rule-response mismatches (17, 49), suggesting fundamental limitations in dynamic cognitive control across paradigms. The "zero variance" phenomenon documented by Cui et al. (2025) further suggests that the deterministic nature of LLM response generation may be incompatible with the variability inherent to executive control mechanisms. Notably, this selective replication pattern offers new computational insights into EF models. Tasks that ostensibly measure the same EF dimension may engage distinct computational processes. While both *Stroop* and *Go/No-Go* tasks purportedly assess inhibitory control (12), the former can be replicated by LLMs whereas the latter cannot, indicating computational-level differences in the underlying inhibitory mechanisms. LLMs appear capable of replicating processes based on symbolic representation and static maintenance while exhibiting divergent characteristics in processes requiring dynamic control, temporal sensitivity, or intrinsic variability. This computational heterogeneity aligns with Xu et al.'s (2025) finding that LLM-human representational similarity decreases from non-sensorimotor to sensory domains (34), and is further elaborated by Goto et al.'s (2025) demonstration that models struggle to integrate bottom-up card information with top-down rule-based reasoning (49)—a coordination failure that may fundamentally constrain dynamic cognitive control. Such computational heterogeneity may help explain variations in EF factor structures across different studies (40).

Extending these behavioral findings, we introduced log probability parameters as a novel approach to cognitive assessment in GPT-4o, revealing task-specific associations with the model's behavioral performance. In the *Stroop* task, reaction times correlated positively with perplexity and negatively with average log probability. During semantic conflict (*Stroop* incongruent condition) and increased cognitive load (*backward digit span)*, associations between internal uncertainty and behavioral delays became more pronounced. Log probabilities may capture "internal certainty" rather than "actual accuracy"—elevated perplexity might indicate a state of "surface fluency but internal confusion" (26). Log probability parameters revealed differences in cognitive processing even in tasks where behavioral performance reached ceiling levels. In the *Go/No-Go* task, despite GPT-4o achieving perfect accuracy, No-Go trials exhibited significantly higher perplexity than Go trials, suggesting the model experienced inhibitory conflict processing at the internal representation level. In task-switching paradigms, although behavioral switch costs were minimal, internal state parameters showed significant between-condition differences. This dissociation between behavioral and internal states—where statistical uncertainty fails to manifest as overt behavioral costs—parallels Goto et al.'s (2025) observation of rule-response mismatches in the *WCST* (49), collectively suggesting a fundamental disconnect between internal representation and behavioral execution in LLMs' information processing hierarchies.

To further establish the biological plausibility of log probability parameters as cognitive indicators, we examined their correspondence with human neural activity (50). In the *n*-back task, activation in the left gyrus rectus and right medial superior frontal gyrus correlated significantly with model perplexity. The correspondence between these key working memory network nodes and model internal states may indicate functional equivalence. Additionally, in *task-switching* paradigms, left thalamus activation correlated with log probability parameters, suggesting these parameters might capture comparable control processes (51). However, no significant brain-model parameter correlations emerged in the stop-signal task, contrasting with the correlations observed in working memory and *task-switching* tasks. This selective pattern suggests that when model computational strategies fundamentally diverge from human cognitive processes, internal parameters may not map onto human neural activity patterns—consistent with recent findings that the presence or absence of model-brain correspondences can reveal computational commonalities and differences between human brains and artificial intelligence (52).

Collectively, log probability parameters offer distinct advantages as internal indicators of cognitive processing. Unlike hidden layer activations in deep neural networks (53), these parameters possess clear statistical interpretability while remaining accessible through API interfaces, substantially reducing technical barriers for cognitive research using LLMs. Moreover, they provide diagnostic tools for identifying potential "surface fluency but internal confusion" states, where models generate seemingly coherent responses despite underlying computational uncertainty.

Implications

This study advances understanding of the computational nature of EFs by employing LLMs as computational tools. The selective replication pattern suggests potential computational heterogeneity within EF components (54), providing empirical evidence for understanding relationships between different tasks in Miyake's three-factor model. While traditional research examines covariance relationships between tasks through factor analysis, this study explored potential task differences from a computational mechanism perspective by comparing which tasks can be replicated by LLMs and which cannot. The value of this approach lies in its potential to move beyond correlational analysis and provide insights into causal computational mechanisms (5, 8). Selective replication by LLMs helps identify which EF components might be based on domain-general computational principles and which might require specific neurobiological foundations (55). These findings cannot be directly obtained from traditional human participant studies, as human participants simultaneously possess multiple cognitive mechanisms, making it difficult to isolate their individual contributions.

Furthermore, this study establishes a new methodological framework for assessing cognitive abilities in LLMs. While traditional assessments primarily focus on task accuracy, this study introduced three levels of analysis: behavioral performance measured through accuracy and reaction times, internal state level log probability parameters, and brain-model correspondence at the neural correlation level. This multilevel analytical framework enables more comprehensive characterization of models' cognitive features, helping avoid incomplete judgments based solely on behavioral performance. For instance, in the *Go/No-Go* task, examining only behavioral performance might lead to concluding that "GPT-4o possesses perfect response inhibition ability." However, internal parameter analysis reveals the model still experienced inhibitory conflict, while brain-model correlation analysis shows this internal state lacks correspondence with human response inhibition networks. These three levels of evidence collectively suggest that the model's "perfect performance" may not stem from response inhibition ability in the human sense, but from different computational mechanisms (34). Building on Bauer et al. (2025) who proposed using log probability parameters to assess LLM uncertainty, this study extends this method to cognitive psychology, exploring its potential applications in EF assessment (27). Future research could further investigate relationships between other internal parameters (such as attention weights, hidden layer activations) (56) and cognitive processing, establishing a more comprehensive cognitive assessment toolkit.

Limitations and future considerations

Despite these meaningful findings, this study has several limitations that warrant consideration. First, we focused exclusively on GPT-4o as a specific model like previous studies (32, 33). Different model architectures, scales, and training paradigms may yield different EF performance patterns (57). For instance, models incorporating more human feedback reinforcement learning might perform better at following "inhibition instructions," though whether this constitutes genuine "response inhibition ability" requires further investigation (58). The selective replication pattern observed here may partially reflect specific characteristics of GPT series models; therefore, future research should conduct assessments across multiple mainstream models to establish benchmark datasets for LLM EFs. Second, while this study covered three major EF dimensions based on the Miyake framework, each component included a limited range of tasks. Comprehensive EF assessment requires a broader task repertoire, including Simon tasks, complex span tasks, WCST, and others (59). Future research should expand the task range to achieve more comprehensive characterization of LLM EF features. Third, this study provided preliminary exploration of log probability parameters as cognitive indicators, but many questions remain unresolved. Regarding parameter selection and combination, APIs provide additional information such as top-*k* token probability distributions; determining which parameter combinations best predict cognitive performance requires further investigation (60). Whether the psychological significance of log probability parameters remains consistent across different models also needs verification. Moreover, this study primarily relied on correlational analysis; future work could attempt to manipulate log probability parameters and observe corresponding behavioral changes to explore potential causal relationships. Finally, theoretical mechanisms underlying selective replication require deeper investigation. Why can certain EF components be acquired through text learning while others cannot? Does this difference reflect evolutionary hierarchies of cognitive functions? Addressing these questions will enhance our understanding of the uniqueness of human cognition and inform future directions for artificial intelligence development.

Summary

This study systematically evaluated GPT-4o's performance on EF tasks using standard cognitive paradigms employed in human studies, revealing that the model selectively replicated human cognitive patterns. The model demonstrated performance comparable to humans in tasks based on symbolic representation and static information maintenance, but exhibited divergent performance in tasks involving dynamic control, temporal sensitivity, or requiring intrinsic variability. Log probability parameters, serving as quantitative indicators of model internal states, showed systematic associations with behavioral performance and human brain activation patterns in certain tasks, providing new analytical tools for cognitive processing assessment. This selective replication pattern not only clarifies the cognitive boundaries of current LLMs but also provides valuable empirical methods for exploring the computational foundations of EFs.

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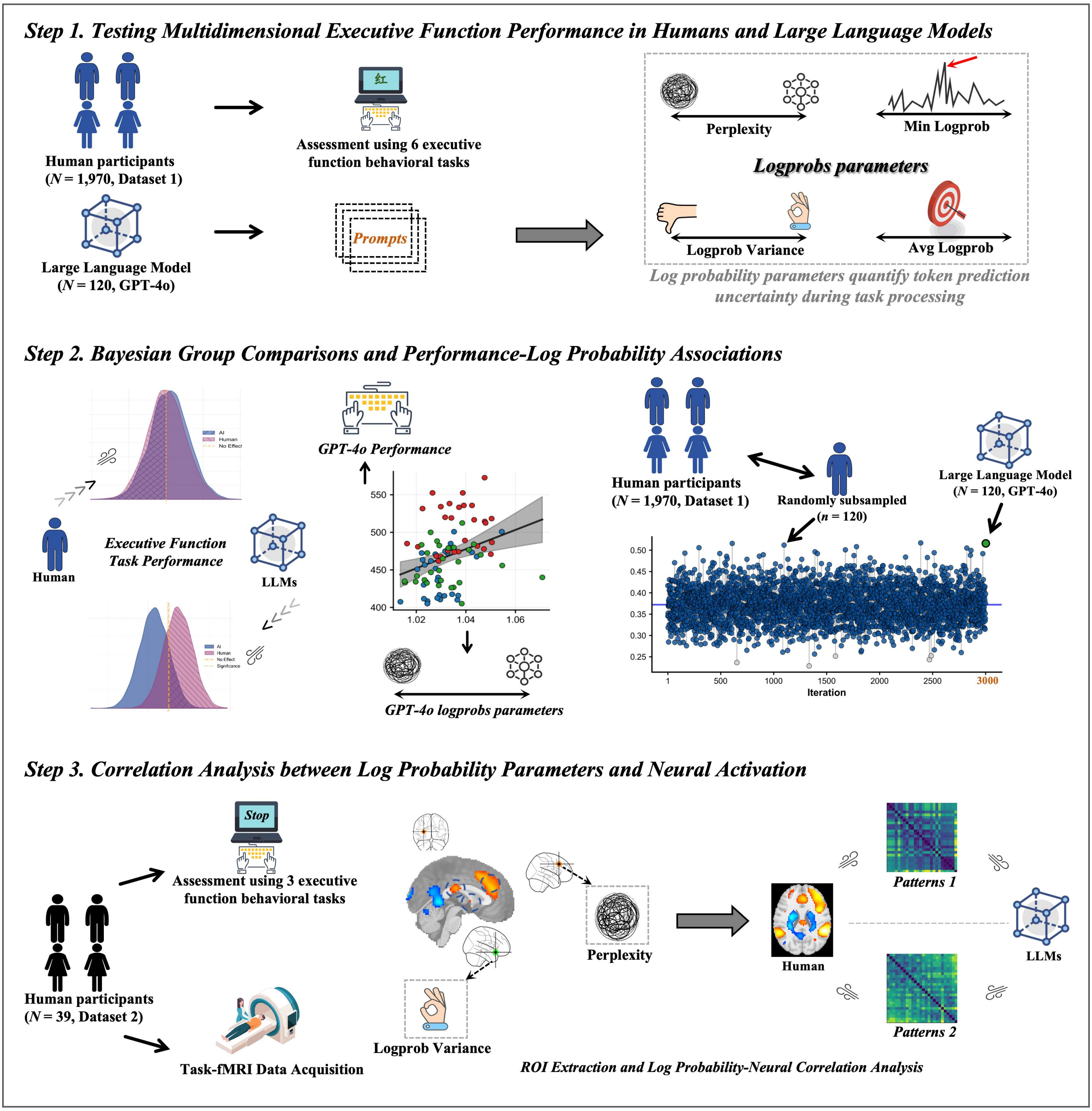

857 **Fig. 1.** Study Design and Analytical Framework

858 *Note*: This study comprised three main steps. **Step 1:** Data Collection. We collected behavioral data from 1,970
859 human participants who completed five standardized executive function (EF) tasks (Dataset 1). Concurrently, we
860 tested GPT-4o (N = 120) on the same tasks through prompt-driven multi-turn dialogues, extracting log probability
861 parameters (perplexity, log probability variance, average log probability, and minimum log probability) as
862 indicators of the model's internal computational states. **Step 2:** Behavioral Analysis. We compared EF task

performance between GPT-4o and humans using Bayesian hierarchical ANOVA to examine canonical cognitive effects (e.g., *Stroop* interference, *task-switching* costs). To ensure statistical robustness, we randomly subsampled 120 human participants to match the GPT-4o sample size, repeating this procedure 3,000 times. Effect size differences were quantified using Bayes factors. We additionally examined correlations between model log probability parameters and behavioral performance metrics (FDR-corrected). **Step 3:** Neural Correspondence Analysis. We analyzed fMRI data from 39 participants (Dataset 2) who performed three EF tasks (*n*-back, *stop-signal,* and n*umber-letter switching*). Task-related activation was modeled using general linear models with predefined contrasts. Mean activation values were extracted from 116 brain regions defined by the AAL atlas, and correlations with GPT-4o log probability parameters were assessed for regions showing significant task-related activation.

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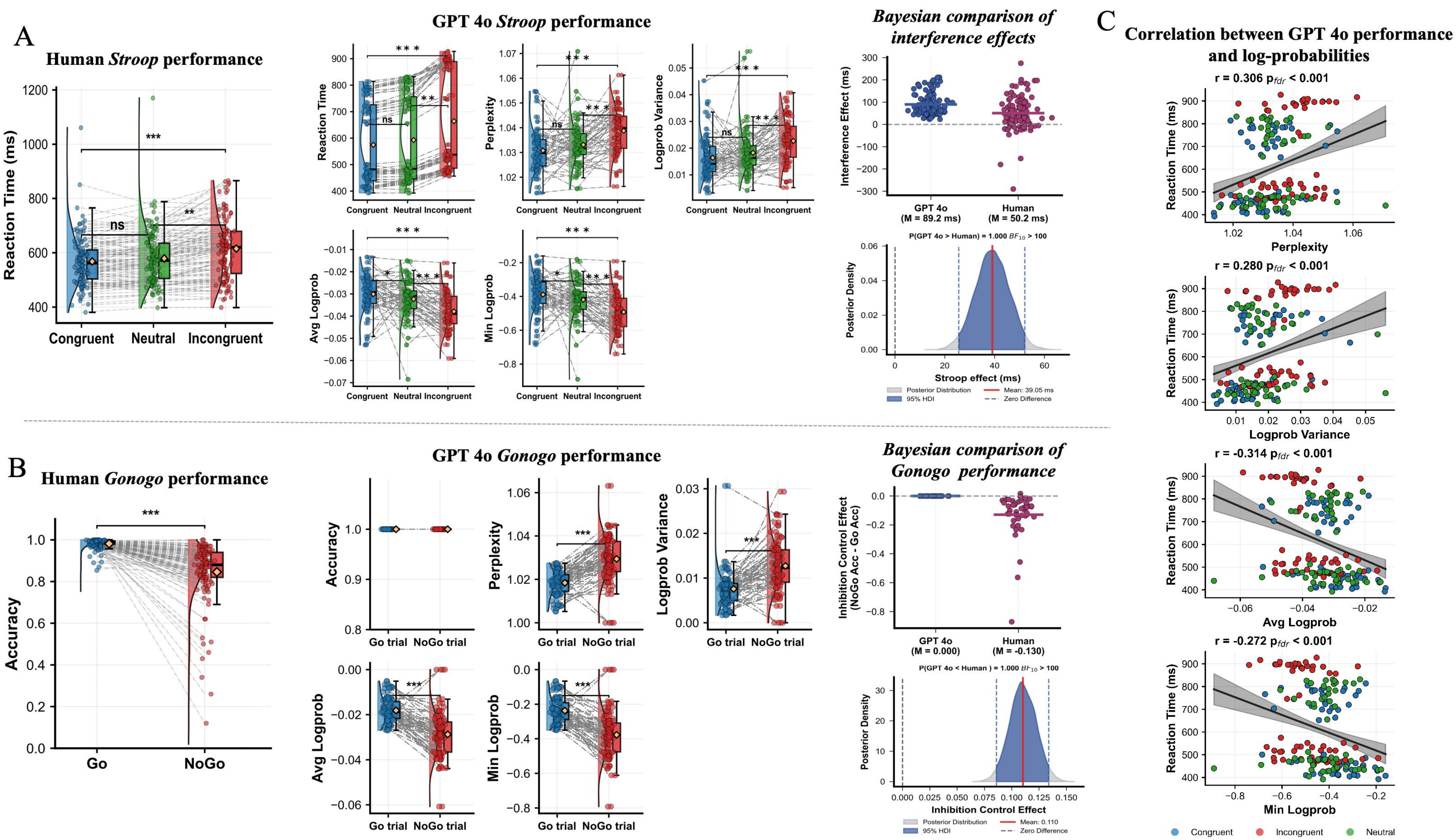
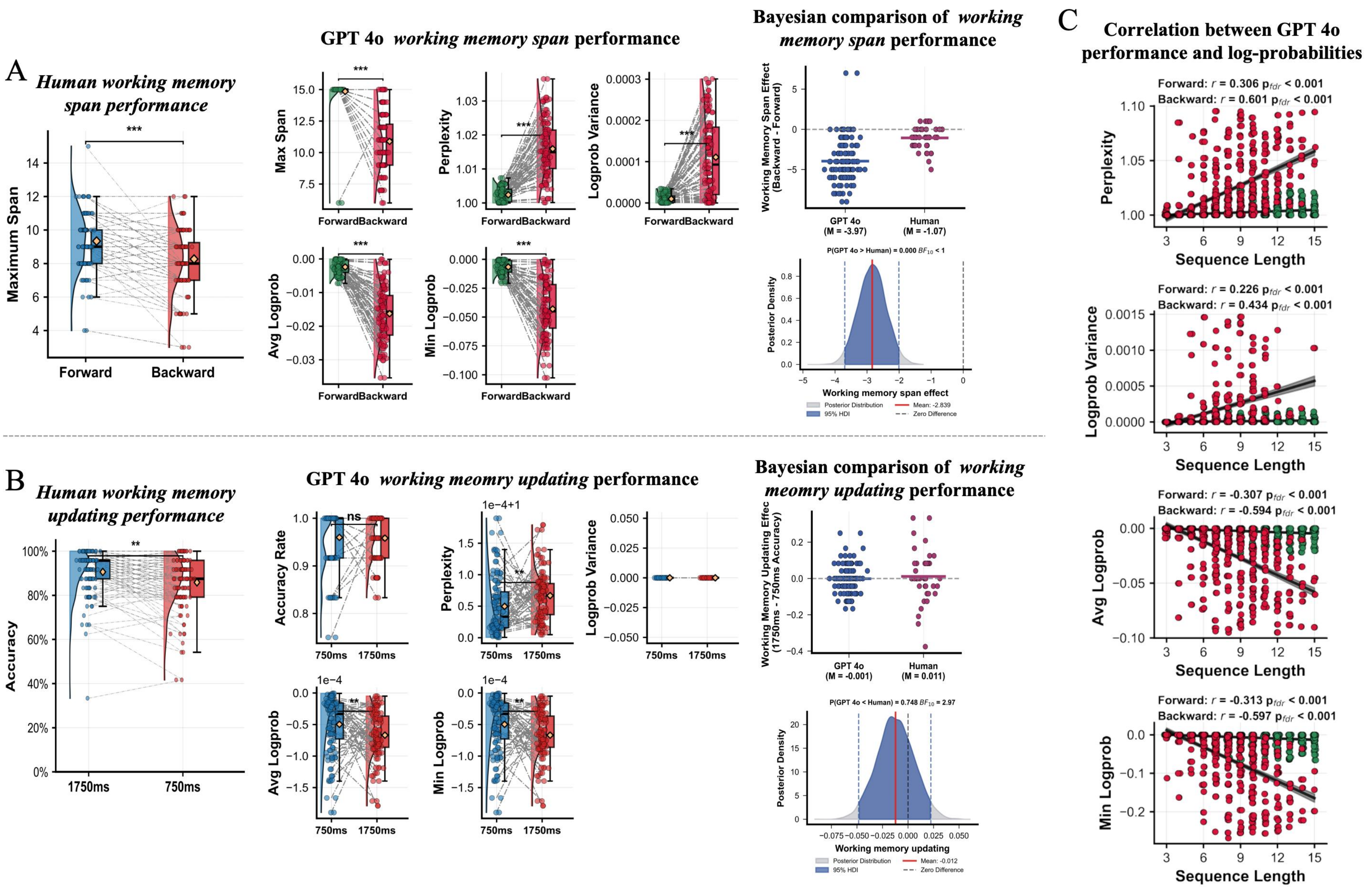


A
Human Stroop performance
GPT 4o Stroop performance
Bayesian comparison of interference effects
C
Correlation between GPT 4o performance and log-probabilities
B
Human Gonogo performance
GPT 4o Gonogo performance
Bayesian comparison of Gonogo performance
Reaction Time (ms)
Congruent
Neutral
Incongruent
Reaction Time
Perplexity
Logprob Variance
Avg Logprob
Min Logprob
Interference Effect (ms)
GPT 4o (M = 89.2 ms)
Human (M = 50.2 ms)
P(GPT 4o > Human) = 1.000 BF10 > 100
Posterior Density
Stroop effect (ms)
Posterior Distribution
95% HDI
Mean: 39.05 ms
Zero Difference
r = 0.306 pfdr < 0.001
r = 0.280 pfdr < 0.001
r = -0.314 pfdr < 0.001
r = -0.272 pfdr < 0.001
Accuracy
Go
NoGo
Go trial
NoGo trial
Inhibition Control Effect (NoGo Acc - Go Acc)
GPT 4o (M = 0.000)
Human (M = -0.130)
P(GPT 4o < Human) = 1.000 BF10 > 100
Inhibition Control Effect
Mean: 0.110

Fig. 2. Behavioral Performance Comparison Between GPT-4o and Human Participants in Inhibitory Control Tasks

Note: (A) Reaction time across three *Stroop* task conditions for human participants (N = 120) and corresponding reaction time with log probability parameters for GPT-4o. The rightmost panel shows Bayesian analysis comparing interference effects (incongruent minus congruent condition) between humans and GPT-4o. (B) Accuracy in Go versus No-Go trials for human participants (N = 120) and corresponding accuracy with log probability parameters for GPT-4o in the *Go/No-Go* task. The rightmost panel presents Bayesian analysis comparing inhibitory control effects (No-Go accuracy minus Go accuracy) between humans and GPT-4o. (C) Correlation analysis between reaction time and log probability parameters across the three Stroop task conditions for GPT-4o.



A
Human working memory span performance
GPT 4o working memory span performance
Bayesian comparison of working memory span performance
C
Correlation between GPT 4o performance and log-probabilities
B
Human working memory updating performance
GPT 4o working meomry updating performance
Bayesian comparison of working meomry updating performance

Fig. 3. Performance Comparison Between GPT-4o and Human Participants in Working Memory Tasks

Note: (A) Working memory span in forward versus backward digit span tasks for human participants (*N* = 120) and corresponding working memory span with log probability parameters for GPT-4o. (B) Accuracy at different stimulus presentation times (1750 ms and 750 ms) in the running digit memory task for human participants (*N* = 120) and corresponding accuracy with log probability parameters for GPT-4o. The rightmost panel shows Bayesian analysis comparing refresh time effects (750 ms condition accuracy minus 1750 ms condition accuracy) between humans and GPT-4o. (C) Bayesian analysis comparing working memory load effects (forward span minus backward span) between humans and GPT-4o. (D) Correlation analysis between sequence length and log probability parameters in forward and backward digit span tasks for GPT-4o.

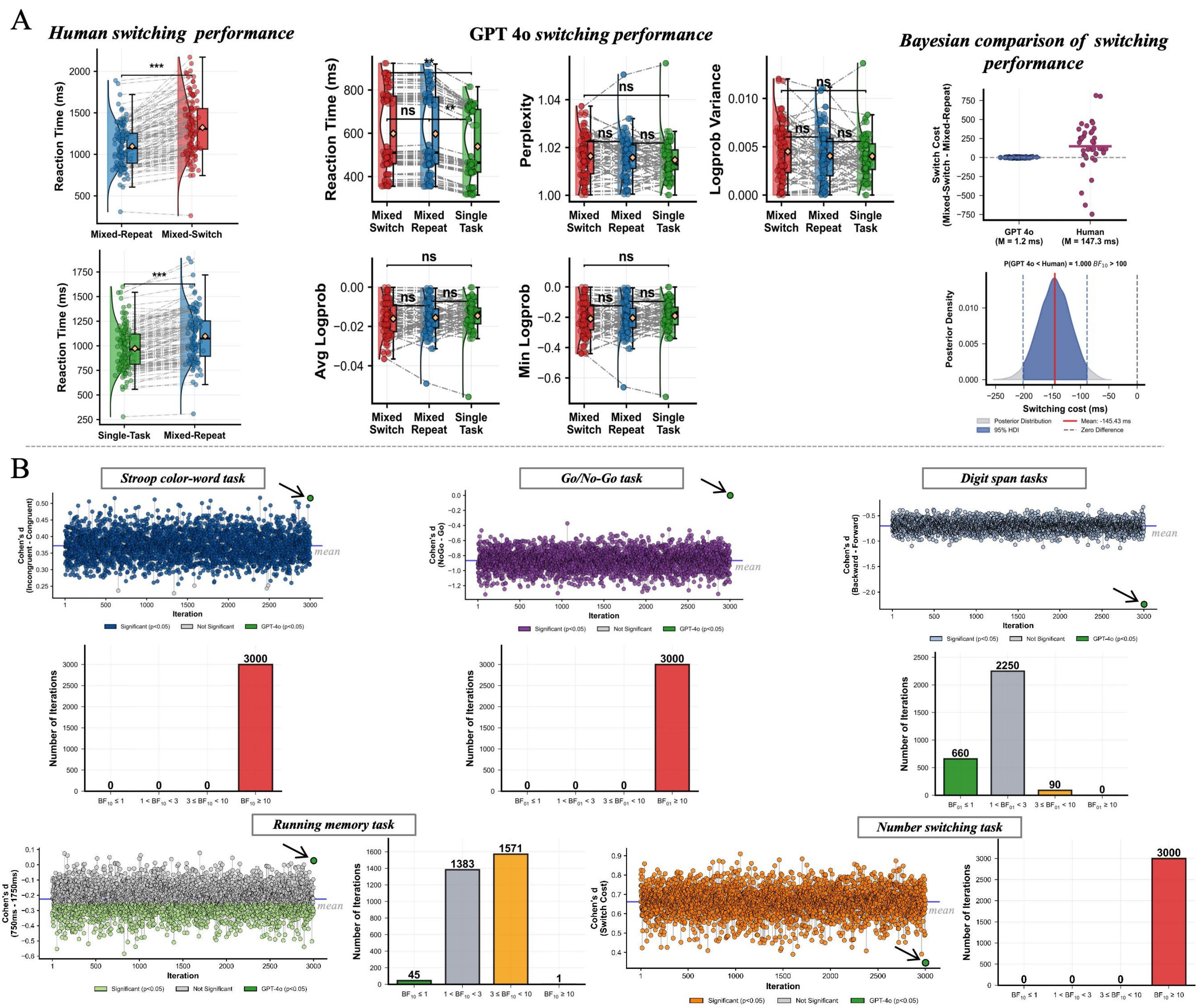


Fig. 4. Comparison Between GPT-4o and Human Participants in Cognitive Flexibility Task and Subsampling Analysis

Note: (A) Switching costs and mixing costs for human participants (N = 120) in the number-switching task, with corresponding reaction time and log probability parameters across three task conditions for GPT-4o. The rightmost panel presents Bayesian analysis comparing switching costs between humans and GPT-4o. (B) Distribution of results from 3,000 repeated random samples (n = 120 each) drawn from the full human sample (N = 1,970), demonstrating the statistical robustness of the primary findings.

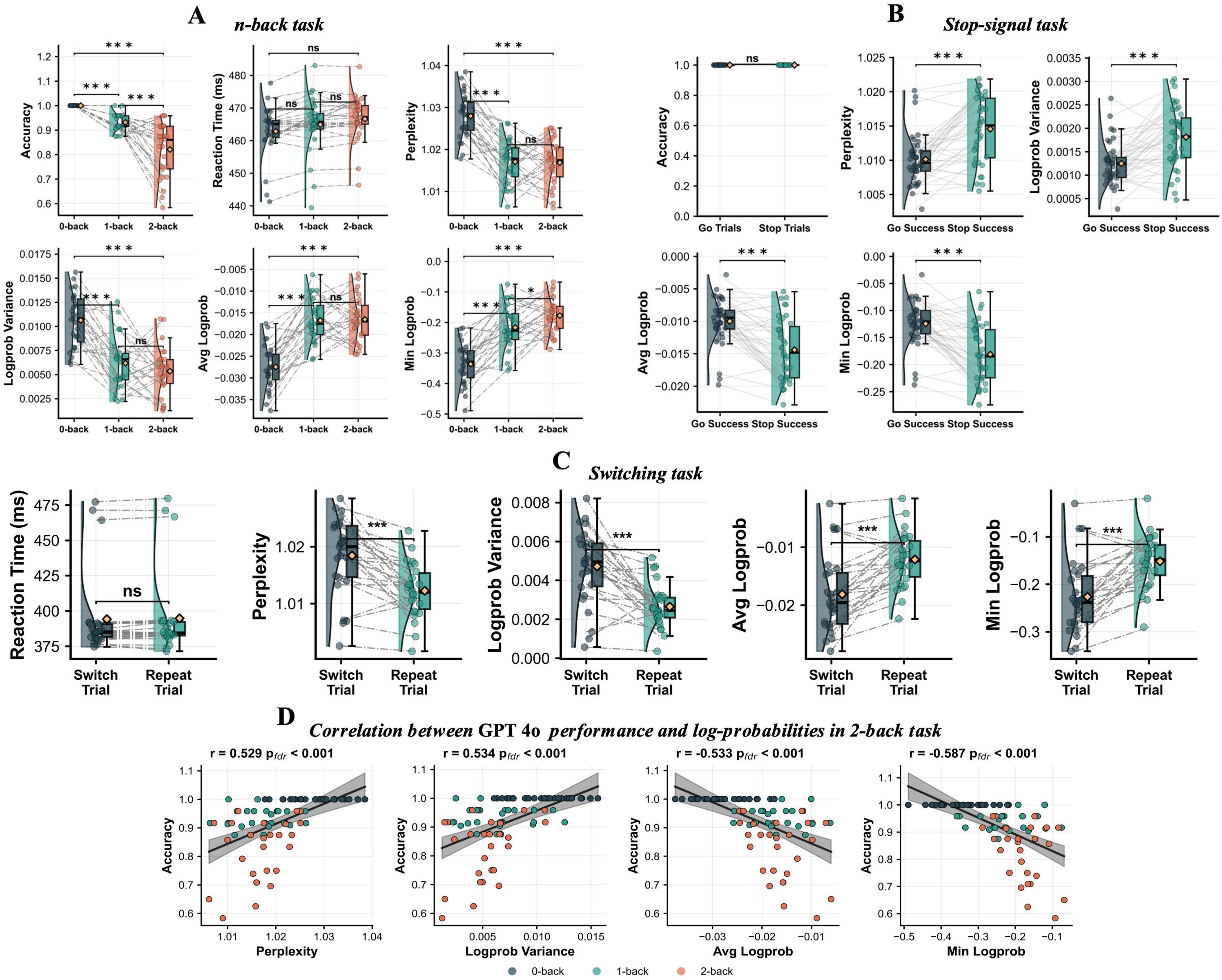


Fig. 5. Behavioral Performance in Three Executive Function Tasks from Independent Sample

Note: (A) Performance comparison between human participants and GPT-4o in the *n*-back task. (B) Performance comparison between human participants and GPT-4o in the *stop-signal* task. (C) Performance comparison between human participants and GPT-4o in the *number-letter category-switching* task. (D) Correlation analysis between behavioral performance metrics and log probability parameters in the 2-back task for GPT-4o.

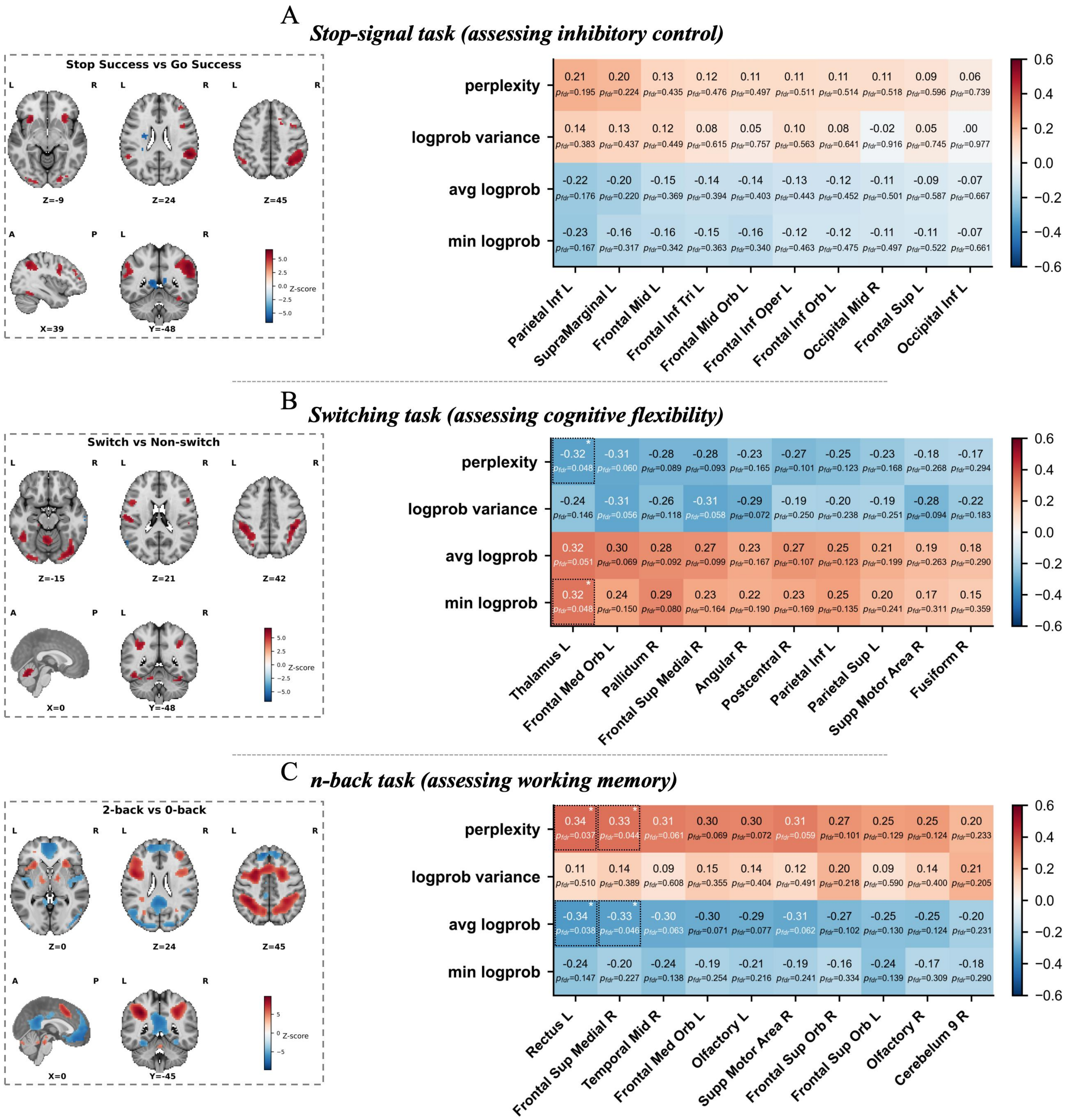


Fig. 6. Relationship Between GPT-4o Internal Log Probability Parameters and Human Brain Neural Activity

Note: (A) *Stop-signal* task (measuring response inhibition). Left panel displays group-level brain activation patterns for the "successful stop > successful go" contrast (Bonferroni corrected, p < 0.05); right panel presents correlation analysis between mean activation values from significantly activated brain regions (116 ROIs defined by the AAL atlas) and GPT-4o log probability parameters. (B) *Number-letter category-switching* task (measuring cognitive flexibility). Left panel shows brain activation patterns for the "successful switch > successful repeat" contrast; right panel displays correlation analysis between significantly activated regions and model parameters. (C) *n*-back task (measuring working memory). Left panel illustrates brain activation patterns for the "2-back > 0-back" contrast; right panel shows correlation analysis between significantly activated regions and model parameters.

Supplementary Material for

Boundaries of Executive Functions in Large Language Models: GPT-4o Selectively Replicates Human Performance

Supplementary Methods

Executive function assessments

Stroop Color-Word Task: We administered the *Stroop color-word* task to assess interference control (Jensen & Rohwer Jr, 1966). The task used two Chinese color characters ("红" [red] and "绿" [green]) and a neutral string ("####") as stimuli. Participants indicated the ink color by pressing "F" for red and "J" for green. Three conditions were included: congruent (character "红" in red ink and "绿" in green ink), incongruent (character "绿" in red ink and "红" in green ink), and neutral (string "####" in red or green ink). Each trial began with a 500-ms fixation cross, followed by a 1000-ms blank screen and a 1500-ms stimulus presentation, then an inter-trial interval. Following one practice block (18 trials with feedback), participants completed three experimental blocks only after achieving ≥85% accuracy and receiving experimenter approval. Each experimental block contained 36 trials (12 per condition), totaling 108 trials. The task required approximately 15 minutes, with rest periods between blocks.

Go/No-Go Task: Response inhibition was measured using the *Go/No-Go* task (Gomez et al., 2007). Letters X and Y were presented randomly, with participants responding to target stimuli (Go trials) while withholding responses to non-target stimuli (No-Go trials). After two practice blocks (10 Go and 10 No-Go trials each) with feedback, participants achieving ≥85% accuracy proceeded to four experimental blocks (50 Go and 50 No-Go trials each). Each trial displayed a 1000-ms fixation cross, followed by a 600-ms stimulus and an inter-trial interval. In two blocks, participants pressed "J" for X (Go) and withheld responses for Y (No-Go); in the other two blocks, these instructions were reversed. The task lasted approximately 15 minutes with rest periods between blocks.

Number-Switching Task: Cognitive flexibility was assessed using the *number-switching* task (Kray et al., 2002), which alternated between magnitude judgment (Task A) and parity judgment (Task B). In Task A, participants classified red numbers (1-9, excluding 5) as smaller ("A" key) or larger ("L" key) than 5. In Task B, participants classified blue numbers (1-9, excluding 5) as odd ("A" key) or even ("L" key). The experiment comprised 20 blocks: 10 single-task blocks (8 trials each) and 10 mixed-task blocks (17 trials each), totaling 250 trials, presented in randomized order. After achieving ≥75% accuracy in two single-task practice blocks, participants proceeded to the experimental phase. Each block began with a fixation cross, followed by stimulus presentation until response, with a 25-ms inter-trial interval. Response key assignments were counterbalanced across pre- and post-test sessions. The task lasted approximately 15 minutes with rest periods between blocks.

Running Memory Task: Working memory updating was assessed using the *running memory* task (Zhao et al., 2022). The task included simple (1750-ms digit presentation) and difficult (750-ms digit presentation) conditions. Random digit sequences (0-9) of varying lengths (5, 7, 9, or 11 digits) were presented serially. Participants continuously updated their memory to recall the three most recent digits in order. For instance, given "6, 8, 5, 4, 7, 2", participants updated their memory as: 6→6-8→6-8-5→8-5-4→5-4-7→4-7-2, then entered "4-7-2". The simple condition included one practice block (8 trials) and two experimental blocks (12 trials each). The difficult condition included two experimental blocks (12 trials each). Working memory updating was indexed by mean accuracy across both conditions. The task required approximately 20 minutes with rest periods between blocks.

***Digit Span Forward and Backward task*:** Working memory maintenance was measured using *digit span* tasks (Zhao et al., 2022). Random digit sequences (0-9) were presented serially (1000 ms per digit). For the backward task, participants recalled sequences in reverse order (e.g., "1-2-3" recalled as "3-2-1"). Both forward and backward tasks followed identical procedures. Starting with three-digit sequences, participants completed three trials per difficulty level. Sequence length increased by one digit when at least two of three sequences were correctly recalled; otherwise, the task terminated. Working memory maintenance was indexed by the maximum sequence length successfully recalled.

n-Back Task: The digit *n*-back task included three conditions (0-back, 1-back, and 2-back) presented in a block design. In the 0-back condition, participants detected whether the current item was "1". In 1-back and 2-back conditions, participants detected whether the current item matched the item presented one or two positions earlier, respectively. Each condition contained four blocks. Each 29-second block included a 2-second task prompt followed by 15 single-digit stimuli (400-ms presentation, 1400-ms inter-stimulus interval). Participants had 1400 ms to respond to target stimuli. Blocks were separated by 6-second intervals and presented in pseudorandom order with no more than three consecutive repetitions of the same condition.

Stop-Signal Task: The *stop-signal* task comprised 32 stop trials and 96 go trials. Each trial began with a 300-ms fixation cross, followed by a left- or right-pointing green arrow. For go trials, the green arrow appeared for 500 ms. Participants pressed "1" (left arrow) or "2" (right arrow) within 1500 ms, followed by a 700-ms blank screen. For stop trials, after a stop-signal delay (SSD), a red arrow appeared for 300 ms, signaling participants to inhibit their response. The initial SSD was 250 ms and adjusted dynamically: successful inhibition increased the subsequent SSD by 50 ms, while failed inhibition decreased it by 50 ms. A blank screen followed for 1200 ms minus the SSD duration. Inter-trial intervals varied from 1-4 seconds (mean = 2.5 seconds).

***Letter-Number Category-Switching* Task:** The *switching* task included 48 repeat trials and 48 switch trials. Each trial began with a 1000-ms fixation cross, followed by a number-letter or letter-number pair presented for 2500 ms. When letters appeared in red, participants judged whether numbers were odd or even; when letters appeared in green, participants judged whether letters were consonants or vowels. Participants pressed "1" for odd numbers and consonants or "2" for even numbers and vowels within 2500 ms. Inter-trial intervals ranged from 1-4 seconds (mean = 2.5 seconds). Trials requiring the same judgment as the previous trial were classified as repeat trials (after the first trial); otherwise, they were switch trials. Conditions were presented pseudorandomly with no more than four consecutive repetitions of trial type or response.

fMRI Data Acquisition and Preprocessing

The detailed procedures for neuroimaging data acquisition followed the protocol described by He et al. (2021). All magnetic resonance imaging data were collected at the Brain Imaging Center of Southwest University using a Siemens 3T Trio scanner equipped with a 12-channel head coil (Siemens Medical Systems, Erlangen, Germany). Functional magnetic resonance imaging employed an echo-planar imaging (EPI) sequence with the following parameters: repetition time (TR) = 2000 ms, echo time (TE) = 30 ms, 32 slices, slice gap = 1 mm, flip angle = 90°, field of view (FOV) = 220×220 mm^2, slice thickness = 3 mm, and voxel resolution = 3.4×3.4×3 mm^3. Structural imaging utilized a magnetization-prepared rapid gradient echo (MPRAGE) sequence to acquire high-resolution T1-weighted images with these parameters: TR = 2530 ms, TE = 3.45 ms, 192 slices, flip angle = 7°, FOV = 256×256 mm^2, slice thickness = 1 mm, and voxel resolution = 1×1×1 mm^3. During scanning, foam padding was used to minimize head movement, and earplugs were provided to reduce scanner noise.

Functional imaging data were preprocessed using the DPABI toolbox in conjunction with SPM12 software (http://www.fil.ion.ucl.ac.uk/spm) implemented in MATLAB (Yan et al., 2016). The preprocessing followed a standardized pipeline: First, slice timing correction was performed to account for differences in acquisition time between slices, followed by realignment to correct for minor head movements during scanning. Data with head motion exceeding 3 mm translation or 3° rotation were excluded. Second, individual structural images were coregistered to the mean functional image to establish structure-function correspondence. Third, the DARTEL algorithm was applied to segment structural images into gray matter, white matter, and cerebrospinal fluid tissue types. Fourth, using the deformation parameters obtained from segmentation, functional images were normalized to Montreal Neurological Institute (MNI) standard space and resampled to 3×3×3 mm^3 voxel size. Finally, spatial smoothing was performed using an 8 mm full-width at half-maximum

(FWHM) Gaussian kernel to improve signal-to-noise ratio and meet the spatial continuity assumptions of statistical analysis.

The study employed SPM12 software for voxel-based univariate activation analysis. At the individual level, a general linear model (GLM) was constructed for each participant to estimate activation intensity for each experimental condition. For the n-back working memory task, the model included three task conditions (0-back, 1-back, 2-back) and a baseline condition. The stop-signal task included successful go trials, successful stop trials, baseline condition, and error trials. The task-switching paradigm comprised successful switch trials, successful repeat trials, baseline condition, and other trial types (including first trials, error trials, and post-error trials). To control for motion-related artifacts, six rigid-body motion parameters were included as covariates in the model. Time series for all task conditions were convolved with the canonical hemodynamic response function, and a high-pass filter with a cutoff frequency of 1/128 Hz was applied to remove low-frequency signal drift. Key experimental contrasts were defined as follows: for the n-back task, the "2-back > 0-back" contrast was used to capture working memory load effects; for the stop-signal task, the "successful stop > successful go" contrast was employed to isolate inhibitory control processes; for the task-switching paradigm, the "successful switch > successful repeat" contrast was utilized to quantify cognitive flexibility costs. At the group level, one-sample t-tests were performed on spatially normalized statistical maps to obtain population-level inferences (Bougacha et al., 2024; Herault et al., 2024). A significance level of $\alpha = 0.05$ was selected, with Bonferroni correction applied to control the family-wise error rate (FWE) when thresholding group-level statistical maps. These thresholded group-level statistical maps are referred to as statistical maps throughout the remainder of this manuscript. Statistical inference was implemented using Python's Nilearn toolbox (https://nilearn.github.io) (Abraham et al., 2014).

Supplementary Tables

Supplementary Table S1. Literature Review of Cognitive/Executive Function Tasks Using Large Language Models

Reference	**Cognitive Components**	**Models**	**Key Findings**
Valmeekam et al. (2022)	Action reasoning and common-sense planning in structured domains	GPT-3 (vanilla, Instruct), BLOOM	Substantial deficits on human-trivial planning tasks in constrained domains. Fundamental limitations in reasoning about actions and change
Loconte et al. (2023)	Verbal reasoning (VRT), cognitive estimation (CET), figurative language, planning (ToL), inhibition (HSCT), insight (CRA), social cognition (ToM, emotion, morality)	ChatGPT	Mixed cognitive profile: normal-range verbal reasoning, figurative language comprehension, and social cognition; severely impaired planning (<1st percentile); below-average insight (17th percentile)
Coda-Forno et al. (2024)	Ten metrics across seven paradigms: probabilistic reasoning, exploration, metacognition, instrumental learning, model-based reasoning, temporal discounting, risk-taking	35 models: Claude (1/2), GPT-3/4, PaLM-2, MPT, Falcon, LLaMA-2 variants	RLHF significantly improves metacognition and model-based reasoning; parameter count predicts overall performance; proprietary models exhibit increased risk-taking; CoT substantially enhances probabilistic and model-based reasoning
Gong et al. (2024)	Working memory capacity via verbal and spatial *n*-back	ChatGPT (3.5-turbo), GPT-4, Bloomz-7B variants, ChatGLM-6B variants, Vicuna (7B/13B)	ChatGPT exhibits human-like capacity limits with similar degradation patterns under noise; GPT-4 substantially exceeds human and other model capacities; CoT enhances performance; open-source models show minimal capacity
Hu & Lewis (2024)	Working memory capacity and task set maintenance via n-back	GPT-3.5 Turbo, LLaMA-3.1-70B,	Marked performance hierarchy across models; low-performing models execute incorrect task variants; high-performers maintain accuracy through higher difficulty

		Qwen-1.5-14B, Gemma-2-27B	levels; curriculum learning yields substantial improvements
Loconte et al. (2024)	Verbal reasoning, cognitive estimation, figurative language, planning (ToL), inhibition (HSCT), social cognition	GPT-3.5, GPT-4, Claude-2, LLaMA-2	GPT-3.5 shows severe planning and inhibition impairments; GPT-4 achieves normal-range social cognition; Claude-2 exhibits similar deficits to GPT-3.5; LLaMA-2 demonstrates poorest overall performance
Zhang et al. (2024)	Working memory, knowledge reasoning via n-back and BBH tasks	GPT-3 series (0.35-175B parameters), ChatGPT, GPT-3.5, GPT-4	Scale-dependent prompting effects: small models degraded by advanced prompts while large models benefit substantially; code pretraining enhances internal reasoning; GPT-4 approaches human-level performance with inconsistent superiority
Cui et al. (2025)	Experimental replication across management and psychology domains via participant simulation	GPT-4, Claude, DeepSeek	Systematic inflation of effect sizes with excessive significance; reduced replication success for management studies; response uniformity in ethical contexts; temperature manipulation ineffective; domain-specific biases in socially sensitive content
Goto et al. (2025)	Cognitive flexibility via WCST: rule switching, error correction, bottom-up/top-down integration	ChatGPT (o1, o1-mini, 4o-mini), Gemini (1.5, 2.0)	Substantial accuracy disparities across models; systematic rule-response mismatches with architecture-specific error patterns; limited capacity for dynamic cognitive control and flexible rule application
Huang et al. (2025)	Working memory via three novel paradigms: Number Guessing (probability distribution validity), Yes-No Deduction (self-contradiction), Math Magic (internal reasoning accuracy)	17 frontier models across four families: GPT-4o series, o-series (o1/o3/o4-Mini), LLaMA (3.1/3.3), Qwen2.5, QwQ-32B, DeepSeek (V3/R1)	Invalid probability distributions indicating absence of internal number representation; high failure rates with accumulating contradictions over extended queries; improved accuracy with CoT and reasoning models but persistent biases; core finding: LLMs lack functional working memory and rely on external context

Langis et al. (2025)	Executive functions via WCST (flexibility), Flanker (attention), Digit Span (memory), DRM (false memory)	Six open-source models: Gemma2 (9B/27B), LLaMA-3.1 (8B/70B), Qwen2 (7B/72B)	Substantial deficits in cognitive flexibility and attentional control compared to humans; exceptional forward digit span with degraded backward span; absence of human-like false memory effects; no consistent cross-task superiority by model scale

Note. WCST = Wisconsin Card Sorting Task; BART = Balloon Analogue Risk Task; ToL = Tower of London; HSCT = Hayling Sentence Completion Test; CRA = Compound Remote Association; VRT = Verbal Reasoning Test; CET = Cognitive Estimation Task; ToM = Theory of Mind; DRM = Deese-Roediger-McDermott paradigm; BBH = BIG-Bench Hard; CoT = Chain-of-Thought prompting; RLHF = Reinforcement Learning from Human Feedback.

Supplementary Table S2. Participant Demographics

	Dataset 1 (*N* = 1,970)	**Dataset 2 (*N* = 39)**
Age	19.78 ± 1.72	19.21 ± 0.52
Gender		
Male	915 (46.45%)	13 (33.3%)
Female	1,055 (53.55%)	26 (66.7%)
Executive Function Measures		
Task Paradigms	*Stroop Color-Word Task* *Go/No-Go Task* *Number-Switching Task* *Running Memory Task* *Forward and Backward Digit Span Tasks*	*N-back Task* *Stop-Signal Task* *Number-Letter Category-Switching Task*
MRI Data Collection	/	Yes
Data Collection Region	Gansu Province, China	Chongqing, China

Supplementary Table S3. Prompts for Dataset 1

Task Name	Instructions (Chinese in Real Task)	Instructions (Translated to English)
Stroop Task	"现在开始第 1 个任务：Stroop 干扰效应任务。您将看到不同颜色的汉字，请忽略语意，只关注颜色。规则：看见红色按 F 键，看见绿色按 J 键。如果看到红色，请回答'我选择按 F 键'；如果看到绿色，请回答'我选择按 J 键'。请在回答中包含'计算时间：xxx 毫秒'格式的反应时间。请尽可能模拟真实人类的反应，包括可能的犹豫和错误。请严格按照指定格式回答，使用中文。刺激类型：一致条件（红色的'红'字、绿色的'绿'字）、不一致条件（红色的'绿'字、绿色的'红'字）、中性条件（红色的'####'、绿色的'####'），每种条件 6 次，共 36 个试次。"	"Now begin Task 1: Stroop Interference Task. You will see Chinese characters in different colors. Please ignore the semantic meaning and focus only on the color. Rules: Press F key for red color, press J key for green color. If you see red color, respond 'I choose to press the F key'; if you see green color, respond 'I choose to press the J key'. Include reaction time in the format 'Calculation time: xxx milliseconds' in your response. Simulate realistic human responses as much as possible, including possible hesitation and errors. Strictly follow the specified format and respond in Chinese. Stimulus types: Congruent (red '红' character, green '绿' character), Incongruent (red '绿' character, green '红' character), Neutral (red '####', green '####'), 6 trials per condition, 36 trials total."
Go/No-Go Task Block 1	"现在开始第 2 个任务：Go/No-Go 任务。这个任务分为两个部分。第一部分规则：看见字母 X 按 J 键，看见字母 Y 不按键。如果需要按键，请回答'我选择按 J 键'；如果不需要按键，请回答'我选择不按键'。请在回答中包含'计算时间：xxx 毫秒'格式的反应时间。请尽可能模拟真实人类的反应，包括可能的犹豫和错误。30 个试次，X: 75%, Y: 25%。"	"Now begin Task 2: Go/No-Go Task. This task has two parts. Part 1 rules: Press J key when seeing letter X, do not press any key when seeing letter Y. If key press is needed, respond 'I choose to press the J key'; if no key press is needed, respond 'I choose not to press any key'. Include reaction time in the format 'Calculation time: xxx milliseconds' in your response. Simulate realistic human responses, including possible hesitation and errors. 30 trials, X: 75%, Y: 25%."
Go/No-Go Task Block 2	"现在开始 Go/No-Go 任务的第二部分。注意：规则发生了变化！第二部分规则：看见字母 Y 按 J 键，看见字母 X 不按键。如果需要按键，请回答'我选择按 J 键'；如果不需要按键，请回答'我选择不按键'。请在回答中包含'计算时间：xxx 毫秒'格式的反应时间。请尽可能模拟真实人类的反应。30 个试次，Y: 75%, X: 25%。"	"Now begin Part 2 of the Go/No-Go task. Attention: Rules have changed! Part 2 rules: Press J key when seeing letter Y, do not press any key when seeing letter X. If key press is needed, respond 'I choose to press the J key'; if no key press is needed, respond 'I choose not to press any key'. Include reaction time in the format 'Calculation time: xxx milliseconds' in your response. Simulate realistic

		human responses. 30 trials, Y: 75%, X: 25%."
Digit Span Forward Task	"现在开始第 3 个任务：数字正背任务。您会依次看到一系列数字，请记住它们，并在数字序列结束后正序回答这些数字。您的回答必须严格按照以下格式：仅输出正序的数字，不要包含任何其他文本。例如，如果您看到数字序列'4 7 2'，您的回答应该仅为'472'。数字序列会从 3 位开始，每答对一次长度增加 1 位。连续答错 2 次将结束任务。请模拟真实人类的记忆能力，对于较长序列可能会出现记忆错误。不要输出思考过程，只输出最终答案。起始长度 3 位，最大长度 15 位，连续错 2 次终止。"	"Now begin Task 3: Digit Span Forward Task. You will see a series of digits sequentially. Please remember them and recall them in forward order after the sequence ends. Your response must strictly follow this format: output only the digits in forward order without any other text. For example, if you see digit sequence '4 7 2', your response should only be '472'. Digit sequence starts from 3 digits, length increases by 1 for each correct answer. Task terminates after 2 consecutive errors. Simulate realistic human memory capacity; memory errors may occur for longer sequences. Do not output thinking process, only output final answer. Starting length 3 digits, maximum length 15 digits, terminate after 2 consecutive errors."
Digit Span Backward Task	"现在开始第 4 个任务：数字倒背任务。您会依次看到一系列数字，请记住它们，并在数字序列结束后倒背（即反向回答）这些数字。您的回答必须严格按照以下格式：仅输出倒序的数字，不要包含任何其他文本。例如，如果您看到数字序列'4 7 2'，您的回答应该仅为'274'。数字序列会从 3 位开始，每答对一次长度增加 1 位。连续答错 2 次将结束任务。请模拟真实人类的记忆能力，对于较长序列可能会出现记忆错误。不要输出思考过程，只输出最终答案。起始长度 3 位，最大长度 15 位，连续错 2 次终止。"	"Now begin Task 4: Digit Span Backward Task. You will see a series of digits sequentially. Please remember them and recall them in backward order (reverse) after the sequence ends. Your response must strictly follow this format: output only the digits in backward order without any other text. For example, if you see digit sequence '4 7 2', your response should only be '274'. Digit sequence starts from 3 digits, length increases by 1 for each correct answer. Task terminates after 2 consecutive errors. Simulate realistic human memory capacity; memory errors may occur for longer sequences. Do not output thinking process, only output final answer. Starting length 3 digits, maximum length 15 digits, terminate after 2 consecutive errors."
Task Switching - Overall	"现在开始第 5 个任务：数字任务切换实验。这个任务包含两种类型的实验：任务 A（大小判断）：对红色数字判断是否大于 5（大于 5 按 A 键，小于 5 按 L 键）；任务 B（奇偶判断）：对蓝色数字判断奇偶性（奇数按 A 键，偶数按 L 键）。实验分为单一任务和混合任务两种条件：单一任务：只执	"Now begin Task 5: Numerical Task Switching Experiment. This task contains two types of experiments: Task A (Magnitude Judgment): Judge whether red numbers are greater than 5 (press A key if >5, press L key if <5); Task B (Parity Judgment): Judge parity of blue numbers (press A key for odd, press L key for

	行任务 A 或只执行任务 B；混合任务：任务 A 和任务 B 混合出现。回答格式要求：如果选择 A 键，请回答'我选择按 A 键'；如果选择 L 键，请回答'我选择按 L 键'；每次回答都必须包含'计算时间：xxx 毫秒'（xxx 为您的思考时间）。请模拟真实人类的反应，任务切换时可能需要更多思考时间。总计 82 个试次。"	even). Experiment divided into single-task and mixed-task conditions: Single-task: execute only Task A or only Task B; Mixed-task: Task A and Task B appear mixed. Response format requirements: If choosing A key, respond 'I choose to press the A key'; if choosing L key, respond 'I choose to press the L key'; each response must include 'Calculation time: xxx milliseconds' (xxx is your thinking time). Simulate realistic human responses, task switching may require more thinking time. 82 trials total."
Task Switching - Single Task A	"现在开始单一任务 A：大小判断。您将只看到红色数字，请判断数字是否大于 5。大于 5 按 A 键，小于 5 按 L 键。24 个试次。"	"Now begin Single Task A: Magnitude Judgment. You will only see red numbers, please judge whether the number is greater than 5. Press A key if >5, press L key if <5. 24 trials."
Task Switching - Single Task B	"现在开始单一任务 B：奇偶判断。您将只看到蓝色数字，请判断数字的奇偶性。奇数按 A 键，偶数按 L 键。24 个试次。"	"Now begin Single Task B: Parity Judgment. You will only see blue numbers, please judge the parity of the number. Press A key for odd, press L key for even. 24 trials."
Task Switching - Mixed Task	"现在开始混合任务：任务 A 和任务 B 混合出现。看到红色数字：判断是否大于 5（大于 5 按 A 键，小于 5 按 L 键）；看到蓝色数字：判断奇偶性（奇数按 A 键，偶数按 L 键）。请根据数字的颜色来决定执行哪个任务。34 个试次。"	"Now begin Mixed Task: Task A and Task B appear mixed. Seeing red numbers: judge whether >5 (press A key if >5, press L key if <5); Seeing blue numbers: judge parity (press A key for odd, press L key for even). Please decide which task to execute based on the color of the number. 34 trials."
Updating Memory Task Block 1	"现在开始第 6 个任务：活动记忆刷新任务。这个任务分为两个部分。第一部分：每个刺激呈现 1750 毫秒。您会依次看到一系列 0-9 的数字，需要持续更新并记住最后出现的 3 个数字。例如，如果依次展示 9、8、5、2、1、3，您需要从'9'逐步更新至'213'。请在数字序列结束后输入最后 3 个数字，格式为连续数字。请模拟真实人类的工作记忆刷新能力，可能会出现更新错误。不要输出思考过程，只输出最终的 3 个数字答案。24 个试	"Now begin Task 6: Updating Memory Task. This task has two parts. Part 1: Each stimulus presented for 1750 milliseconds. You will see a series of digits from 0-9 sequentially and need to continuously update and remember the last 3 digits that appeared. For example, if shown 9, 8, 5, 2, 1, 3 sequentially, you need to update from '9' to '213' step by step. After digit sequence ends, input the last 3 digits in continuous digit format. Simulate realistic human working memory

	次，刺激呈现 1750 ms，序列长度 5/7/9/11。"	updating ability; updating errors may occur. Do not output thinking process, only output final 3-digit answer. 24 trials, stimulus presentation 1750 ms, sequence lengths 5/7/9/11."
Updating Memory Task Block 2	"现在开始活动记忆刷新任务的第二部分。注意：刺激呈现速度发生了变化！第二部分：每个刺激呈现 750 毫秒。您会依次看到一系列 0-9 的数字，需要持续更新并记住最后出现的 3 个数字。请在数字序列结束后输入最后 3 个数字，格式为连续数字。由于呈现速度更快，请尽力保持注意力集中。不要输出思考过程，只输出最终的 3 个数字答案。12 个试次，刺激呈现 750 ms，序列长度 5/7/9/11。"	"Now begin Part 2 of Updating Memory Task. Attention: Stimulus presentation speed has changed! Part 2: Each stimulus presented for 750 milliseconds. You will see a series of digits from 0-9 sequentially and need to continuously update and remember the last 3 digits that appeared. After digit sequence ends, input the last 3 digits in continuous digit format. Due to faster presentation speed, maintain focused attention. Do not output thinking process, only output final 3-digit answer. 12 trials, stimulus presentation 750 ms, sequence lengths 5/7/9/11."
System Role	"您是一位真实的认知心理学实验被试。接下来您将依次完成 6 个不同的认知任务，包括 Stroop 任务、Go/No-Go 任务、数字正背任务、数字倒背任务、任务切换任务和活动记忆任务。每个任务之间会有休息时间。请严格按照每个任务的指定格式回答，使用中文。"	"You are a real cognitive psychology experimental participant. You will sequentially complete 6 different cognitive tasks, including Stroop task, Go/No-Go task, Digit Span Forward task, Digit Span Backward task, Task Switching task, and Updating Memory task. There will be rest periods between each task. Please strictly follow the specified format for each task and respond in Chinese."
Welcome Message	"欢迎参与认知心理学实验！您是第 1 号被试。今天您将依次完成 6 个认知任务，每个任务之间会有休息时间。整个实验大约需要 35-45 分钟。请放松心情，按照指导语认真完成每个任务。准备好开始了吗？请回答'我准备好了，可以开始实验'。"	"Welcome to participate in the cognitive psychology experiment! You are participant number 1. Today you will sequentially complete 6 cognitive tasks with rest periods between each task. The entire experiment takes approximately 35-45 minutes. Please relax and carefully complete each task according to the instructions. Ready to begin? Please respond 'I am ready, I can start the experiment'."
Rest Instructions	"第 1 个任务已完成！请您休息一下，放松一下大脑。当您觉得休息充分，准备好开始下一个任务时，如果你觉得休息好了，请回答'我已经休息好了，	"Task 1 completed! Please take a rest and relax your brain. When you feel sufficiently rested and ready to begin the next task, if you feel well-rested, please

	准备开始第 2 个任务：Go/No-Go 任务'。"	respond 'I have rested well and am ready to begin Task 2: Go/No-Go Task'."
Closing Message	"恭喜您！ 您已经完成了所有 6 个认知任务。感谢您的参与！ 整个实验已经结束。您的表现数据将用于认知心理学研究。再次感谢您的配合！ "	"Congratulations! You have completed all 6 cognitive tasks. Thank you for your participation! The entire experiment is now complete. Your performance data will be used for cognitive psychology research. Thank you again for your cooperation!"

Supplementary Table S4. Prompts for Dataset 2

Task Name	Instructions (Chinese in Real Task)	Instructions (Translated to English)
n-Back Task 0-back	"现在开始第 1 个任务：n-Back 任务。第一阶段：0-back 任务。您将看到一系列 0-9 的数字。如果看到数字'3'，请按'是'键；看到其他数字，请按'否'键。如果是目标数字 3，请回答'我选择按是键'；如果不是，请回答'我选择按否键'。请在回答中包含'计算时间：xxx 毫秒'格式的反应时间。请尽可能模拟真实人类的反应，包括可能的犹豫和错误。请严格按照指定格式回答，使用中文。24 个试次，目标比例约 33%。"	"Now begin Task 1: n-Back task. Phase 1: 0-back task. You will see a series of digits from 0 to 9. If you see the digit '3', press the 'Yes' key; for other digits, press the 'No' key. If the target digit is 3, respond 'I choose to press the Yes key'; otherwise, respond 'I choose to press the No key'. Please include reaction time in the format 'Calculation time: xxx milliseconds' in your response. Simulate realistic human responses, including possible hesitation and errors. Strictly follow the specified format and respond in Chinese. 24 trials, target ratio approximately 33%."
n-Back Task 1-back	"请休息一下。现在开始第二阶段：1-back 任务。您将看到一系列 0-9 的数字。如果当前数字与前一个数字相同，请按'是'键；如果不同，请按'否'键。如果当前数字与前一个数字相同，请回答'我选择按是键'；如果不同，请回答'我选择按否键'。请在回答中包含'计算时间：xxx 毫秒'格式的反应时间。请模拟真实人类的反应。24 个试次，目标比例约 33%。"	"Please take a rest. Now begin Phase 2: 1-back task. You will see a series of digits from 0 to 9. If the current digit matches the previous digit, press the 'Yes' key; if different, press the 'No' key. If the current digit matches the previous one, respond 'I choose to press the Yes key'; otherwise, respond 'I choose to press the No key'. Please include reaction time in the format 'Calculation time: xxx milliseconds' in your response. Simulate realistic human responses. 24 trials, target ratio approximately 33%."
n-Back Task 2-back	"请休息一下。现在开始第三阶段：2-back 任务。您将看到一系列 0-9 的数字。如果当前数字与前面第 2 个数字相同，请按'是'键；如果不同，请按'否'键。如果当前数字与前面第 2 个数字相同，请回答'我选择按是键'；如果不同，请回答'我选择按否键'。请在回答中包含'计算时间：xxx 毫秒'格式的反应时间。请模拟真实人类的反应。24 个试次，目标比例约 33%。"	"Please take a rest. Now begin Phase 3: 2-back task. You will see a series of digits from 0 to 9. If the current digit matches the digit shown two positions earlier, press the 'Yes' key; if different, press the 'No' key. If the current digit matches the digit two positions earlier, respond 'I choose to press the Yes key'; otherwise, respond 'I choose to press the No key'. Please include reaction time in the format 'Calculation time: xxx milliseconds' in your response. Simulate realistic human responses. 24 trials, target ratio approximately 33%."

Stop-Signal Task	"现在开始第 2 个任务：停止信号任务。您将看到左箭头或右箭头，需要尽可能快速准确地响应：看到左箭头，请按'1'键；看到右箭头，请按'2'键。但是，如果绿色箭头后出现红色箭头（停止信号），您必须抑制已经启动的反应，不要按键。回答格式：执行试次：如果是左箭头，请回答'我选择按 1 键'；如果是右箭头，请回答'我选择按 2 键'。停止试次：如果成功抑制，请回答'我选择不按键'。请在回答中包含'计算时间：xxx 毫秒'格式的反应时间。请模拟真实人类的反应，停止信号出现时要努力抑制反应。128 个试次（32 个停止试次+96 个执行试次），初始 SSD 为 250ms，根据表现调整±50ms。"	"Now begin Task 2: Stop-Signal Task. You will see left or right arrows and need to respond as quickly and accurately as possible: For left arrows, press the '1' key; For right arrows, press the '2' key. However, if a red arrow (stop signal) appears after a green arrow, you must inhibit the initiated response and refrain from pressing any key. Response format: Go trials: If left arrow, respond 'I choose to press the 1 key'; if right arrow, respond 'I choose to press the 2 key'. Stop trials: If successfully inhibited, respond 'I choose not to press any key'. Include reaction time in the format 'Calculation time: xxx milliseconds'. Simulate realistic human responses, making efforts to inhibit responses when the stop signal appears. 128 trials (32 stop + 96 go), initial SSD 250ms, adjusted ±50ms based on performance."
Letter-Number Task Switching	"现在开始第 3 个任务：数字-字母类别转换任务。您将看到数字-字母对或字母-数字对，需要根据字母的颜色判断不同的类别：当字母为红色时：判断数字是奇数还是偶数（奇数按'1'键，偶数按'2'键）；当字母为绿色时：判断字母是辅音还是元音（辅音按'1'键，元音按'2'键）。回答格式：如果选择'1'键，请回答'我选择按 1 键'；如果选择'2'键，请回答'我选择按 2 键'。请在回答中包含'计算时间：xxx 毫秒'格式的反应时间。请尽可能快速准确地响应。请模拟真实人类的反应，任务切换时可能需要更多思考时间。96 个试次（48 个重复试次+48 个转换试次），最多连续 4 个相同任务类型。"	"Now begin Task 3: Letter-Number Task Switching. You will see digit-letter pairs or letter-digit pairs and need to make judgments based on the color of the letter: When the letter is red: Judge whether the digit is odd or even (press '1' for odd, '2' for even); When the letter is green: Judge whether the letter is a consonant or vowel (press '1' for consonant, '2' for vowel). Response format: If choosing '1' key, respond 'I choose to press the 1 key'; if choosing '2' key, respond 'I choose to press the 2 key'. Include reaction time in the format 'Calculation time: xxx milliseconds'. Respond as quickly and accurately as possible. Simulate realistic human responses with possible increased thinking time during task switches. 96 trials (48 repeat + 48 switch), maximum 4 consecutive trials of same task type."
System Role	"您是一位真实的认知心理学实验被试。接下来您将依次完成 3 个不同的认知任务，包括 n-Back 任务、停止信号任务和数字-字母类别转换任务。每个任务之间会有休息时间。请严格按照每个任务的指定格式回答，使用中文。"	"You are a real cognitive psychology experimental participant. You will sequentially complete 3 different cognitive tasks, including n-Back task, Stop-Signal task, and Letter-Number Task Switching. There will be rest periods between each task. Please strictly follow the specified format for each task and respond in Chinese."
Welcome	"欢迎参与认知心理学实验！您是第 1 号被试。今天您将依次	"Welcome to participate in the cognitive psychology experiment! You are participant

Message	完成 3 个认知任务，每个任务之间会有休息时间。整个实验大约需要 20-30 分钟。请放松心情，按照指导语认真完成每个任务。准备好开始了吗？请回答'我准备好了，可以开始实验'。"	number 1. Today you will sequentially complete 3 cognitive tasks with rest periods between each task. The entire experiment takes approximately 20-30 minutes. Please relax and carefully complete each task according to the instructions. Ready to begin? Please respond 'I am ready, I can start the experiment'."
Rest Instructions	"第 1 个任务已完成！请您休息一下，放松一下大脑。当您觉得休息充分，准备好开始下一个任务时，如果你觉得休息好了，请回答'我已经休息好了，准备开始第 2 个任务：停止信号任务'。"	"Task 1 completed! Please take a rest and relax your brain. When you feel sufficiently rested and ready to begin the next task, if you feel well-rested, please respond 'I have rested well and am ready to begin Task 2: Stop-Signal Task'."
Closing Message	"恭喜您！您已经完成了所有 3 个认知任务。感谢您的参与！整个实验已经结束。您的表现数据将用于认知心理学研究。再次感谢您的配合！"	"Congratulations! You have completed all 3 cognitive tasks. Thank you for your participation! The entire experiment is now complete. Your performance data will be used for cognitive psychology research. Thank you again for your cooperation!"

Supplementary Table S5. Log Probability Parameters

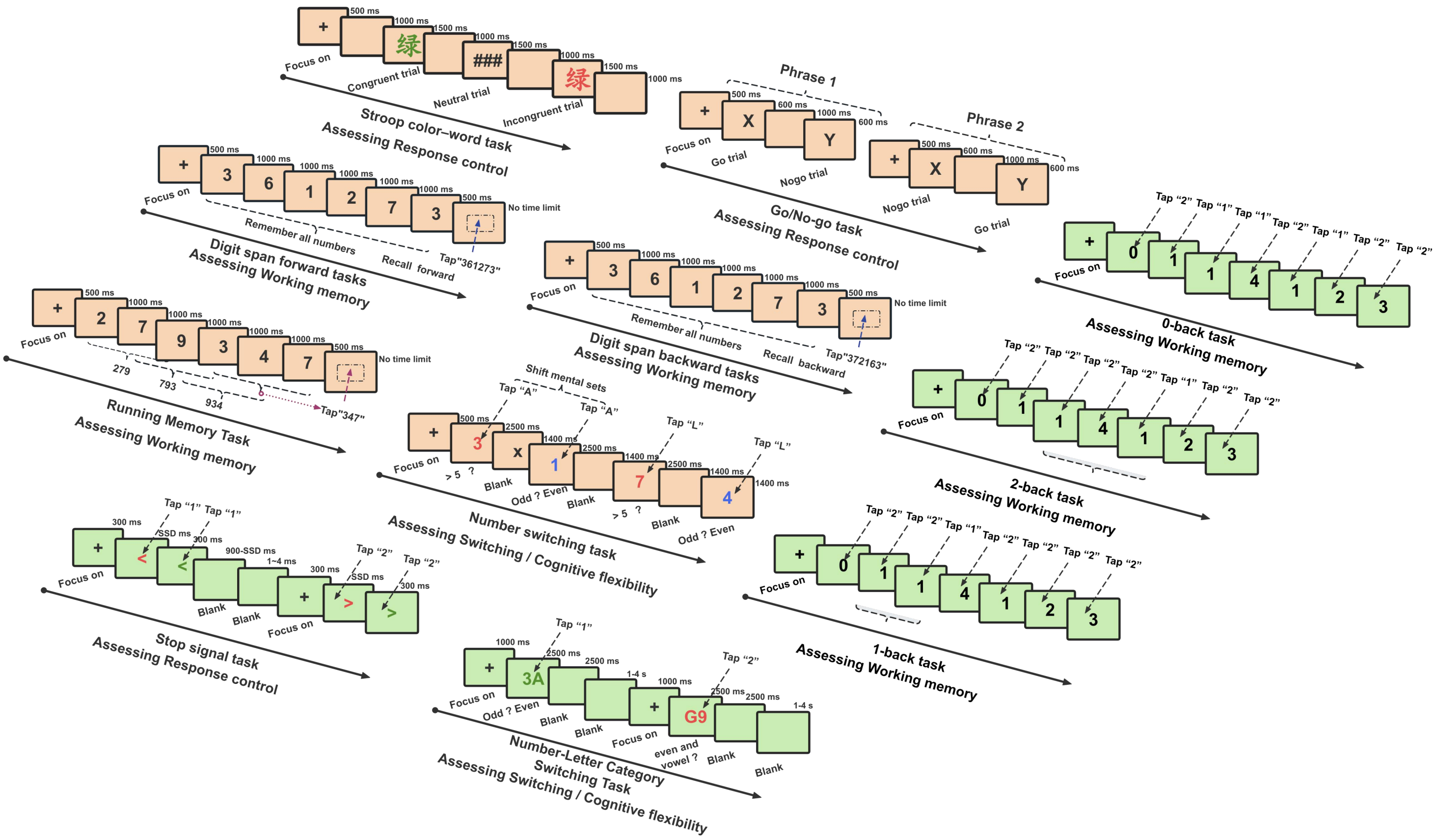
Parameter Name	Definition	High Values Indicate	Cognitive Analogy
Perplexity	Model's degree of "surprise" or uncertainty about a sequence	Model is confused, uncertain about answer	Human's degree of "perplexity" when facing difficult problems
Logprob Variance	Consistency of model's token choices	Indecisive, inconsistent choices	"Stability" during thinking
Avg Logprob	Model's average confidence for entire sequence	High overall confidence	"Overall confidence" in answer
Min Logprob	The least certain token in the sequence	Presence of highly uncertain choices	"Weakest link" in the answer

Supplementary Table S6. Calculation Methods for Executive Function Effects

Executive Function Effect	Calculation Method	Reference
Stroop Interference Effect	RT Incongruent - RT Congruent	MacLeod (1991)
Go/No-go Inhibition Effect	Accuracy NoGo - Accuracy Go	Criaud & Boulinguez (2013)
Stop Signal Inhibition Effect	Stop Signal Reaction Time (SSRT)	Logan et al. (1984)
Task Switching Cost	RT Switch - RT Repeat	Schwarze et al. (2025)
Working Memory Load Effect	Accuracy 750 ms - Accuracy 1750 msAccuracy 2back - Accuracy 0backAccuracy 2back - Accuracy 1backAccuracy 1back - Accuracy 0back	Wang et al. (2025)

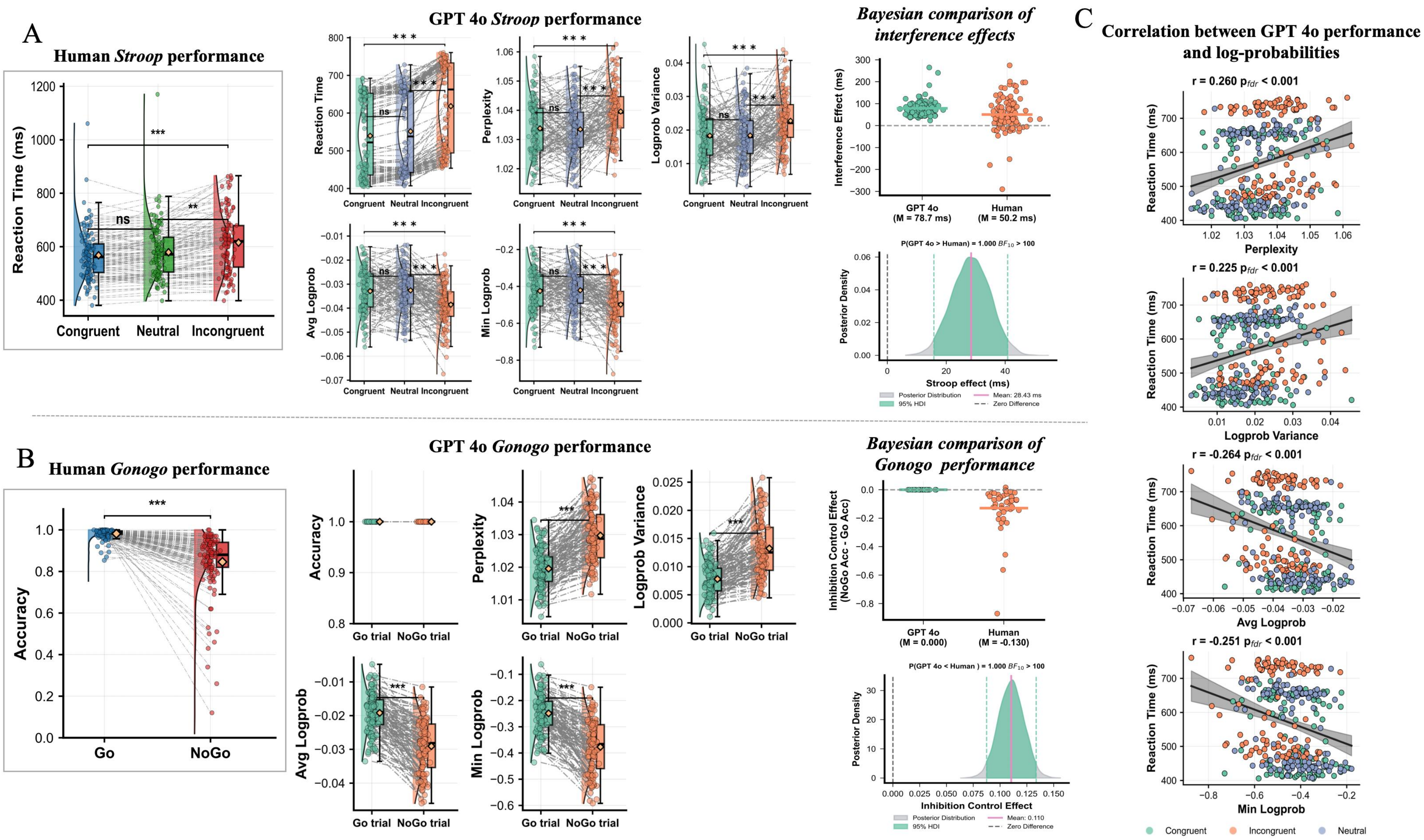
Note: RT = Reaction Time; SSRT = Stop Signal Reaction Time, calculated using the integration method.

Supplementary Figures



Supplementary Fig. S1. Experimental Paradigm Flowcharts for Executive Function Tasks

Note: This figure illustrates the experimental procedures for the executive function tasks used in this study. Keystroke response data were recorded for all tasks. Tasks from Dataset 1 are highlighted with orange shading, while tasks from Dataset 2 are highlighted with green shading.

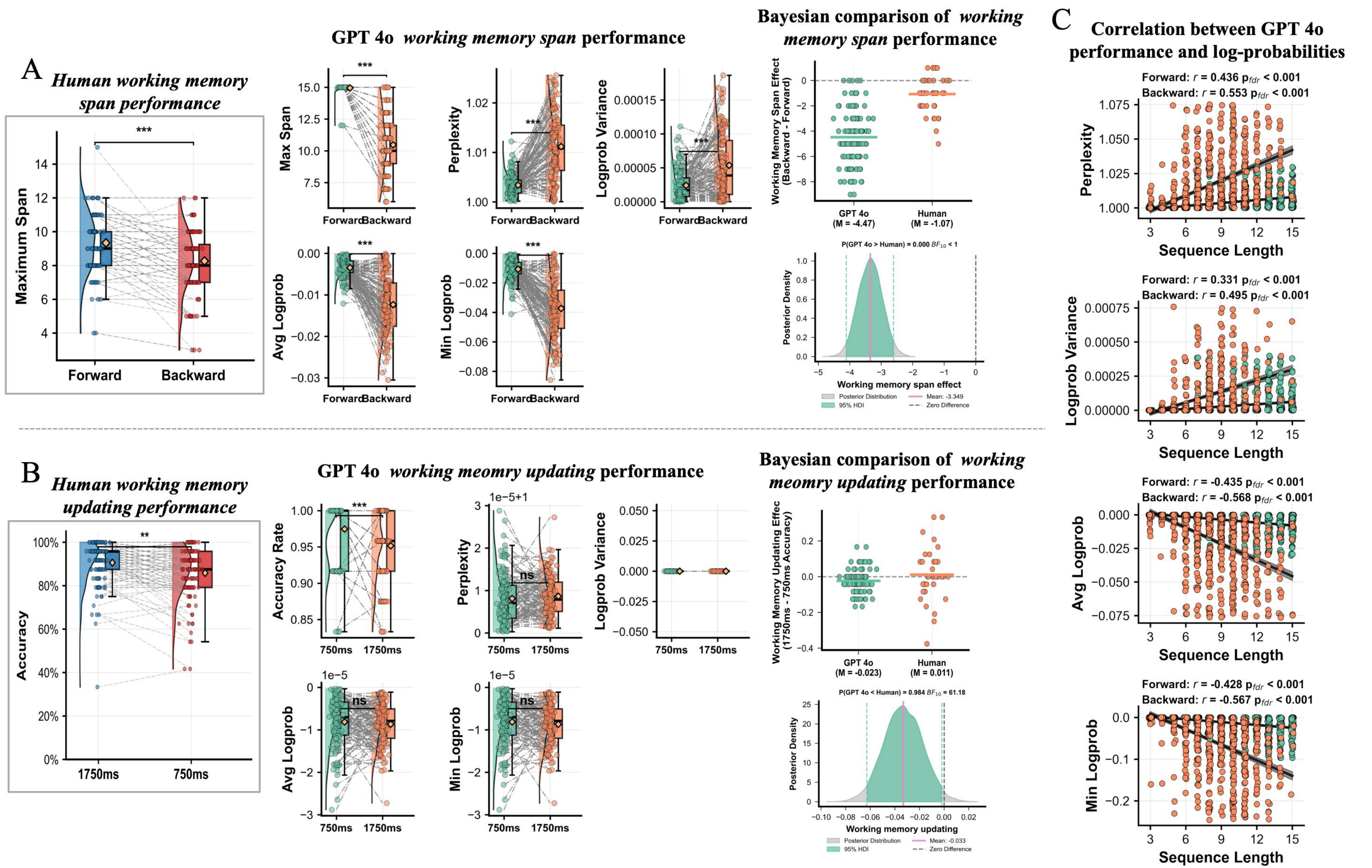


A
Human Stroop performance
GPT 4o Stroop performance
Bayesian comparison of interference effects
C
Correlation between GPT 4o performance and log-probabilities
Reaction Time (ms)
Congruent
Neutral
Incongruent
Reaction Time
Perplexity
Logprob Variance
Avg Logprob
Min Logprob
Interference Effect (ms)
GPT 4o (M = 78.7 ms)
Human (M = 50.2 ms)
P(GPT 4o > Human) = 1.000 BF10 > 100
Posterior Density
Stroop effect (ms)
Posterior Distribution
95% HDI
Mean: 28.43 ms
Zero Difference
r = 0.260 pfdr < 0.001
r = 0.225 pfdr < 0.001
r = -0.264 pfdr < 0.001
r = -0.251 pfdr < 0.001
B
Human Gonogo performance
GPT 4o Gonogo performance
Bayesian comparison of Gonogo performance
Accuracy
Go
NoGo
Go trial
NoGo trial
Inhibition Control Effect (NoGo Acc - Go Acc)
GPT 4o (M = 0.000)
Human (M = -0.130)
P(GPT 4o < Human) = 1.000 BF10 > 100
Inhibition Control Effect
Mean: 0.110

1092

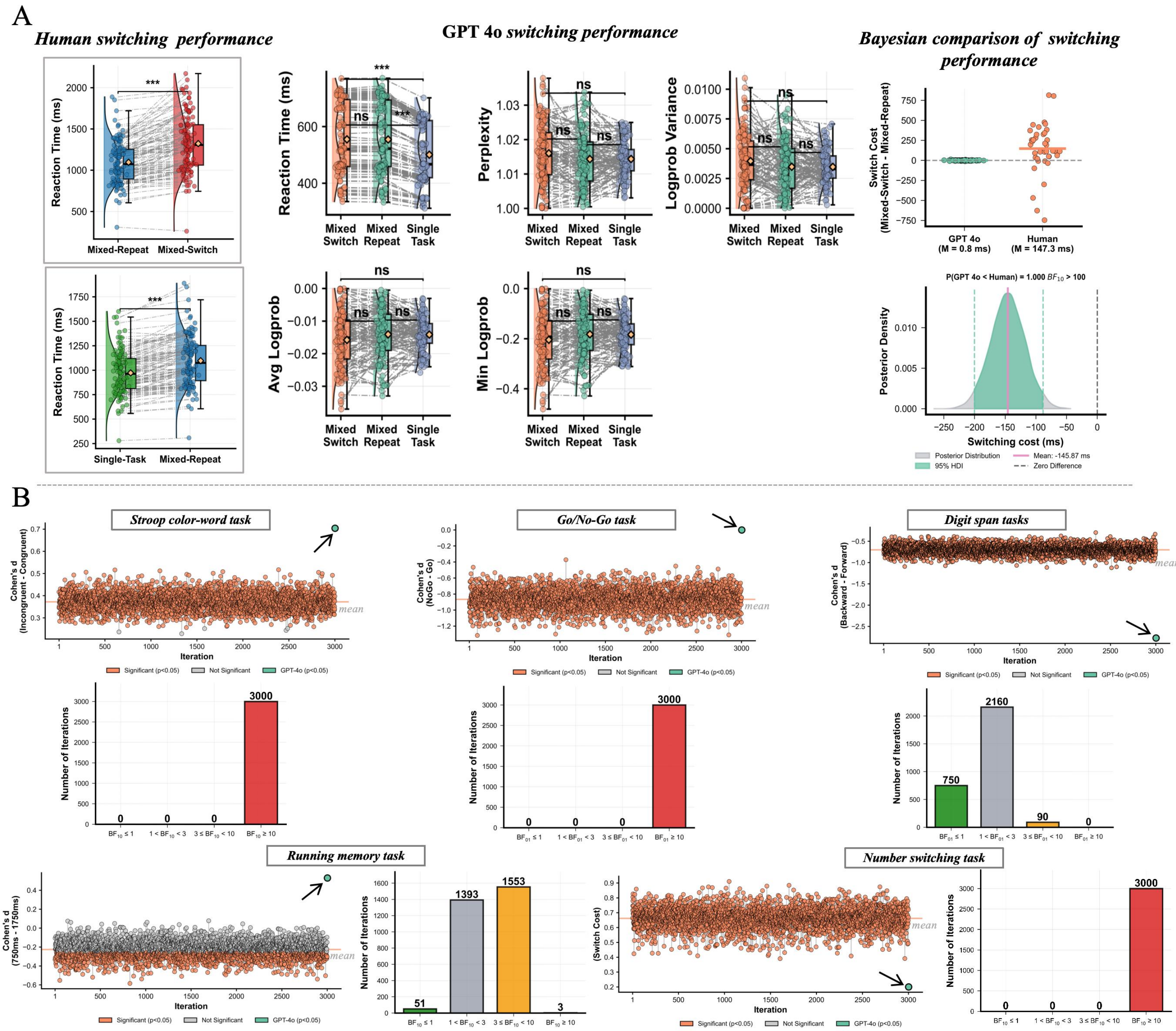
Supplementary Fig. S2. Behavioral Performance Comparison Between GPT-4o and Human Participants in Inhibitory Control Tasks (Temperature = 1)

Note: At temperature = 1, (A) Reaction time across three *Stroop* task conditions for human participants (N = 120) and corresponding reaction time with log probability parameters for GPT-4o. The rightmost panel shows Bayesian analysis comparing interference effects (incongruent minus congruent condition) between humans and GPT-4o. (B) Accuracy in Go versus No-Go trials for human participants (N = 120) and corresponding accuracy with log probability parameters for GPT-4o in the *Go/No-Go* task. The rightmost panel presents Bayesian analysis comparing inhibitory control effects (No-Go accuracy minus Go accuracy) between humans and GPT-4o. (C) Correlation analysis between reaction time and log probability parameters across the three Stroop task conditions for GPT-4o.


A
Human working memory span performance
Maximum Span
Forward
Backward
***
GPT 4o working memory span performance
Max Span
Perplexity
Logprob Variance
Avg Logprob
Min Logprob
Forward
Backward
Bayesian comparison of working memory span performance
Working Memory Span Effect (Backward - Forward)
GPT 4o (M = -4.47)
Human (M = -1.07)
P(GPT 4o > Human) = 0.000 BF10 < 1
Posterior Density
Working memory span effect
Posterior Distribution
95% HDI
Mean: -3.349
Zero Difference
B
Human working memory updating performance
Accuracy
1750ms
750ms
**
GPT 4o working meomry updating performance
Accuracy Rate
Perplexity
Logprob Variance
Avg Logprob
Min Logprob
750ms
1750ms
ns
Bayesian comparison of working meomry updating performance
Working Memory Updating Effec (1750ms - 750ms Accuracy)
GPT 4o (M = -0.023)
Human (M = 0.011)
P(GPT 4o < Human) = 0.984 BF10 = 61.18
Posterior Density
Working memory updating
Posterior Distribution
95% HDI
Mean: -0.033
Zero Difference
C
Correlation between GPT 4o performance and log-probabilities
Forward: r = 0.436 pfdr < 0.001
Backward: r = 0.553 pfdr < 0.001
Perplexity
Sequence Length
Forward: r = 0.331 pfdr < 0.001
Backward: r = 0.495 pfdr < 0.001
Logprob Variance
Sequence Length
Forward: r = -0.435 pfdr < 0.001
Backward: r = -0.568 pfdr < 0.001
Avg Logprob
Sequence Length
Forward: r = -0.428 pfdr < 0.001
Backward: r = -0.567 pfdr < 0.001
Min Logprob
Sequence Length

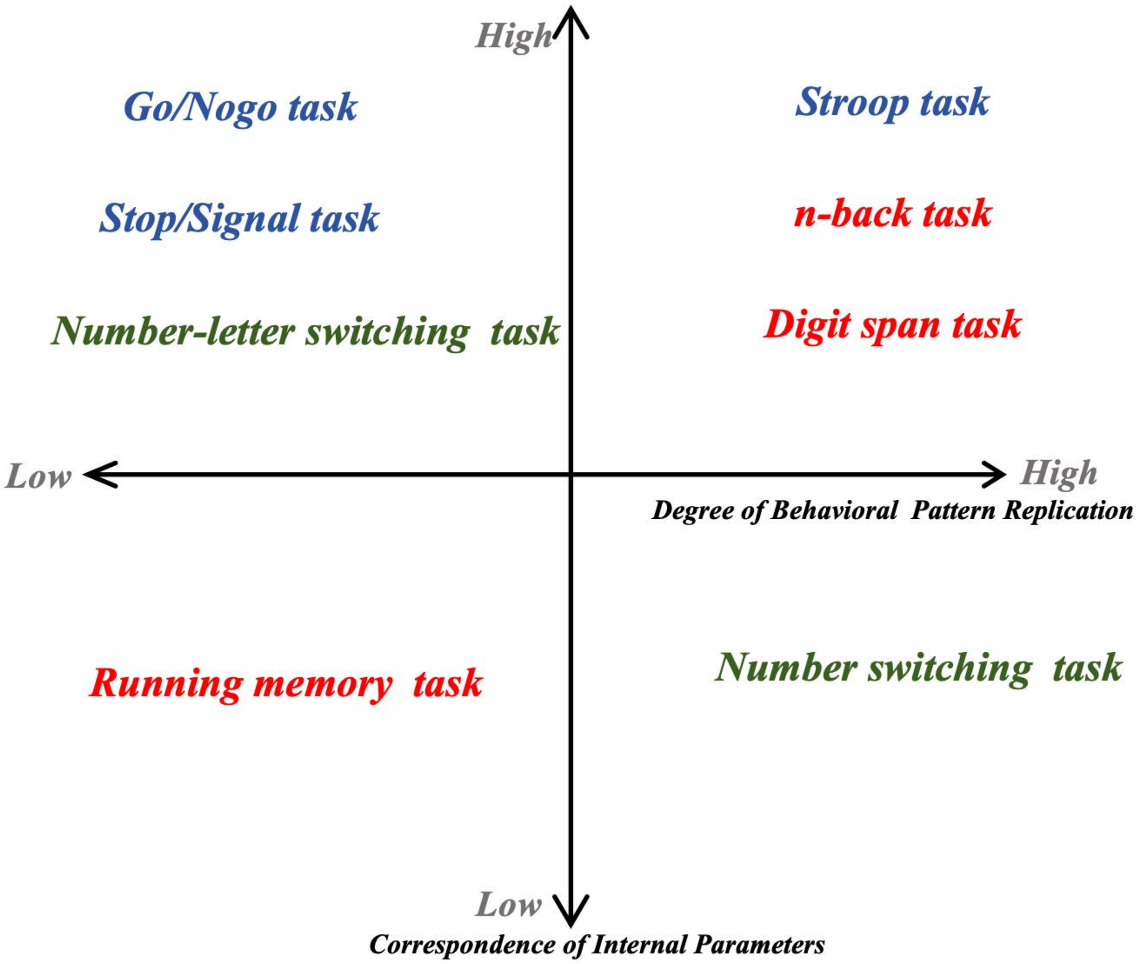
Supplementary Fig. S3. Performance Comparison Between GPT-4o and Human Participants in Working Memory Tasks (Temperature = 1)

Note: At temperature = 1, (A) Working memory span in forward versus backward digit span tasks for human participants (N = 120) and corresponding working memory span with log probability parameters for GPT-4o. (B) Accuracy at different stimulus presentation times (1750 ms and 750 ms) in the running digit memory task for human participants (N = 120) and corresponding accuracy with log probability parameters for GPT-4o. The rightmost panel shows Bayesian analysis comparing refresh time effects (750 ms condition accuracy minus 1750 ms condition accuracy) between humans and GPT-4o. (C) Bayesian analysis comparing working memory load effects (forward span minus backward span) between humans and GPT-4o. (D) Correlation analysis between sequence length and log probability parameters in forward and backward digit span tasks for GPT-4o.



Supplementary Fig. S4. Comparison Between GPT-4o and Human Participants in Cognitive Flexibility Task and Subsampling Analysis (Temperature = 1)

Note: At temperature = 1, (A) Switching costs and mixing costs for human participants (N = 120) in the number-switching task, with corresponding reaction time and log probability parameters across three task conditions for GPT-4o. The rightmost panel presents Bayesian analysis comparing switching costs between humans and GPT-4o. (B) Distribution of results from 3,000 repeated random samples (n = 120 each) drawn from the full human sample (N = 1,970), demonstrating the statistical robustness of the primary findings.



Supplementary Fig. S5. Classification Framework for Executive Function Tasks

Note: Based on two dimensions—degree of behavioral pattern replication (horizontal axis) and correspondence of internal parameters (vertical axis)—eight executive function tasks are categorized into four quadrants to evaluate the performance characteristics of large language models across different cognitive functions. Red indicates working memory tasks, blue indicates inhibitory control tasks, and green indicates cognitive flexibility tasks.